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InTech

COVER STORY

Industrial energy conservation: Where does the reasoning begin?

By Bill Lydon, Chief Editor, *InTech*

Industrial energy conservation has become a big topic. Many technologies can be complex and expensive. Most industrial plants today have "low hanging fruit" opportunities that can readily be identified and will deliver quick payback with small investments.

PROCESS AUTOMATION

16 Processing heavier crudes to meet future needs

By Joseph McMullen, Brittany Doyle, and David Bluck

Crude oil is a non-renewable resource with the overall supply continuously decreasing, making it important to better exploit heavy oils or oil sands which are harder to process. Using a consortium, better modeling is being done to improve designs and operations to obtain these heavy oils or oil sands and refine this lower-quality feedstock.

FACTORY AUTOMATION

20 Small multi-axis industrial robots add new twist to lean manufacturing

By Charlie Miller

New smaller robots are ideal for a range of applications, including assembly, machine tending, pick-and-place, dispensing, and packaging. Robots offer the ultimate in repeatability and have an inherent ability to be changed, adding a valuable dimension of flexibility to the production process.

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By Platt Beltz

Today's low-cost, high-quality PC-based simulators make simulation training affordable for new and experienced operators at all types of facilities.

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28 Uninterruptible power supplies and cybersecurity

By Michael A. Stout

The recent number of cyber-attacks and their level of sophistication have demonstrated the inadequate network security measures employed by many. The critical selection and proper configuration of an Uninterruptible Power Supply (UPS) SNMP/HTTP agent option is vital to network security, but often an afterthought.

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By Allan G. Kern, P.E.

The orifice plate stands the test of time and remains industry's most popular choice for flow measurement. With new sizing rules, orifice plates get an upgrade in improved accuracy and turndown.

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How can knowledge be preserved when baby boomer engineering experts begin to retire? Simple but structured interdisciplinary knowledge containers can help companies to keep, transfer, and apply engineering expertise. Read Carlos Delgado's feature at **www.isa.org/intech/201202web.**

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Getting technical education right

By Bill Lydon, *InTech*, Chief Editor

I recently visited Lee College in Baytown, Tex. (www.lee.edu), which is a community college doing an amazing job of training automation and instrument professionals that are needed by industry. The core of the Lee College mission, "We build and deliver just in time, targeted, and customized training to meet industry's call for new hires and/or incumbents." The college works closely with industry to understand talent needs and builds programs that will meet those needs. It is getting results being ranked sixth in the nation for degrees awarded in science and technology with approximately 50% of Lee College students in technical programs. Lee has a more diverse student population than other community colleges of comparable size, including 46% White, 30% Hispanic, and 18% African-American. Graduates are getting good paying jobs because they are work ready.

First, I was impressed full-time faculty members are required to have a minimum of 15 years' direct industry experience. The experience of the staff is obvious as I talked with them and visited the student labs, where they use real-world industrial instruments configured in working processes. The labs are built to teach full process control with industrial field hardware, including instrumentation, drives, and pumps moving fluids. One instructor talked about how students invariably will ask "what if" questions, and he lets the students try it out in the lab and see the validity or their idea or problems created. He noted when students see the consequences of doing things wrong working in the safe environment of the lab they are likely to remember the lesson.

Lee College has several professional partnerships and affiliations to stay engaged in the community to be sure they are doing the right things. Partners include the International Society of Automation (ISA), National Science Foundation, Department of Labor, and Fieldbus Foundation. It is worth noting they have 96 student members in the school's ISA chapter, putting them in the top five worldwide. (The top two were from China.) Last year, Lee College held an event, "Automation Day" in conjunction with the Automation Federation, with more than 650 high-school students from several local school districts introducing students and teachers to automation and the different fields of study and jobs available.

Chuck Carter, center director, Lee College Center for Digital and Fieldbus Technological Education, provided his thoughts on programs at the college. Carter expressed his belief in the value of hands-on experience, and he championed the building of a full-scale processing facility at the college that has been running since 1998. The college is working to cultivate more internship programs with companies since they have been successful for the students and participating companies.

Carter said perspective students can relate to a number of potential careers, such as firefighter, police, pilot, and auto mechanic, since these bring to mind a visual image, but they have no idea that jobs exist in the automation and control field. The college has been doing outreach to inform the community about opportunities in the field.

Carter said what I believe illustrates the overall attitude at Lee College, "We tell everybody when they ask what they need to come in and take the program to be successful. Number one, you need the want to; if you've got the want to, by golly gee, we have the resources to work with you."

Carter mentioned they are always looking for good equipment to give students hands-on training. If you can help, contact him at ccarter@lee.edu.

The industry needs more cooperation between users and technical schools. Think about how you can help.

Please share your thoughts at blydon@isa.org.

ISA *INTECH* STAFF

CHIEF EDITOR
Bill Lydon
blydon@isa.org

PUBLISHER
Susan Colwell
scolwell@isa.org

ASSOCIATE PRODUCTION EDITOR
Emily Blythe Kovac
ekovac@isa.org

ART DIRECTOR
Colleen Casper
ccasper@isa.org

GRAPHIC DESIGN SPECIALIST
Pam King
pking@isa.org

ISA PRESIDENT
Robert E. Lindeman, CAP, PMP

PUBLICATIONS VICE PRESIDENT
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EDITORIAL ADVISORY BOARD
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Building a strong business case

Thanks for the enlightening article in *InTech* (July/August 2011 "Talk to me"). Nothing seems to change. I was espousing your exact sentiments using **ALL** advantages to sell automation to the decision makers (DMs). The DMs need to know the business case for spending money. I was promoting this thinking as early as the 1970s. Even the DMs with an engineering background could not understand the inherent advantages of automated process systems without a strong business case set before them. It is incumbent upon the controls engineers of today to come out of the closet and learn to develop a strong business case for their automated process systems. It is no longer enough to just be an engineer designing and implementing automated systems; engineers must learn and use the fundamentals of building a business case to sell their project to DMs.

It is no longer enough to just be an engineer designing and implementing automated systems; engineers must learn and use the fundamentals of building a business case to sell their project to DMs.

Can *InTech* run a series of articles about how to build a strong business case for automated systems in today's world? An ongoing column in every issue to re-enforce the concepts of building a business case would be very helpful. Some examples from readers would be very helpful. Getting money to implement automated systems is all but impossible in many stressed industries. Your participation through *InTech* would be a great advantage to ISA members. You might want to think of webinars on the subject, as well.

Bob Giese

Go prepared

I appreciate Bruce Slade's exhortation and tips for "working" a room ("Final Say," July/August *InTech*). I too, like Slade, was very backward in my youth until I moved away from my hometown for work. I undertook a self-study to overcome my strong tendencies to be a "wall-flower," trying to blend into the woodwork. The first thing I did was observe others who seemed to be comfortable mingling in a crowd.

In addition to the tips you have given, I do the following: Go prepared with a pen and 3x5 card in my pocket. Another thing, I remind myself that I was a visitor once, and it is a very uncomfortable feeling. A third thing I do when I'm introduced is to repeat the new person's name, and I concentrate intensely to get the name correct. I try to continue the conversation for two minutes using their name three times while looking them in the eye.

The fourth thing is to write highlights of our conversation on the back side of their business card or the 3x5 card in my pocket. The fifth thing I do is make an introduction to someone else I met earlier. If all else fails, when I see the person later, I take out my 3x5 card and ask them to write down their name and company.

Thanks again for Slade's exhortation. Engineering folks are seldom "minglers" by nature. I've been told on several occasions that I am the exception rather than the rule of the stereotypical engineering type.

Jack R. Jones

Replacing motors

Good article (September/October "Talk to me"). Your comment about switching to "high efficiency" motors only partially solves the problem since most motors are over-sized and not operating at maximum efficiency ... and power factor. So instead of a 10hp motor, try replacing it with a 7.5hp motor, etc.

This is based upon over 30 years of combined experience with Westinghouse, as well as a major IOU, etc., selling capacitors and motor repair as well as DSM—"energy efficiency program" (design/justification/evaluation).

Rick Gordon

Manually verifying every flowmeter wastes time and money. I wish I could just tell which ones need attention.

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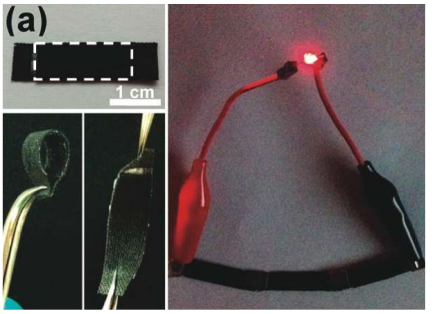
Flexible power source from soot

Electronic devices such as displays, sensors, and medical implants are on their way to becoming flexible. But flexible power sources to operate them are only starting to catch up. Researchers now report a simple method to fabricate bendable supercapacitors that uses carbon nanoparticles from soot, according to *Chemical & Engineering News*.

Supercapacitors, like batteries, store energy. While batteries store and release charge through chemical reactions, supercapacitors store it on the surface of their electrodes. Supercapacitors can charge in minutes instead of hours and can recharge millions of times. Unfortunately, they hold less energy per weight than batteries. To improve supercapacitors' energy density, researchers have replaced activated charcoal electrodes with materials with higher surface area, such as carbon nanotubes and graphene, a one atom-thick sheet of carbon. But making and using these materials typically requires complex, costly methods.



Pliable power
A flexible supercapacitor (left, top) can fold into a loop (left, bottom.) Three such supercapacitors strung in a series can power a light-emitting diode (right).
Source: ACS Nano

Carbon nanoparticles, on the other hand, are cheap and easy to make, says Zhong Lin Wang, a materials science and engineering professor at Georgia Institute of Technology. He thought the nanoparticles could form cheap yet effective supercapacitor electrodes. Wang and his colleagues in China started building such electrodes by holding a flexible carbon substrate in front of an ethanol flame for 30 seconds. The flame deposited a thin layer of carbon nanoparticles, each about 7 nm wide, on the fabric. To make the electrode, the researchers then sputtered manganese oxide nanorods on top of the nanoparticle film. Manganese oxide's high charge storage capacity improves the performance of carbon-based supercapacitor electrodes.

Finally, to make the supercapacitor, the researchers took two of the prepared electrodes, infused them with a polymer gel electrolyte, and sandwiched a standard cellulose separator material between the electrodes.

The resulting foldable device stores 4.8 watt-hours of energy per kg and has a power density of 14 kilowatts per kg. Wang's supercapacitors should be sufficient to run small devices such as sensors and radio-frequency identification tags, Wang said.

Calculator provides ROI for safety automation

Engineers, plant managers, and environmental health and safety (EH&S) professionals now have a tool to calculate the potential annual return they will receive if they invest in an integrated safety automation system. Rockwell Automation developed the free Safety Return on Investment (ROI) tool in partnership with J.B. Titus. The web-based tool addresses manufacturers' need for a tool to help quantify potential savings and productivity gains from new investments in safety.

At its core, the Safety ROI tool relies on a basic calculation: benefits divided by costs equals ROI. The tool combines injury and productivity data and collects input from users in five categories:

- Estimated project amount
- Overall equipment effectiveness, based on increases in machine availability because of reduced unscheduled downtime and increases in manufacturing output
- Increased capital-asset depreciation
- Direct injury costs
- Indirect injury costs (regulatory noncompliance fines/repair costs)

The tool also allows users to adjust the ratio of indirect-to-direct injury costs from 1:1 to 14:1, or to enter zeros for indirect and direct injury costs, based on company requirements.

—News brief courtesy of Automation.com

Mystery of lead-acid battery current solved

Chemists have solved the 150-year-old mystery of what gives the lead-acid battery, found under the hood of most cars, its unique ability to deliver a surge of current.

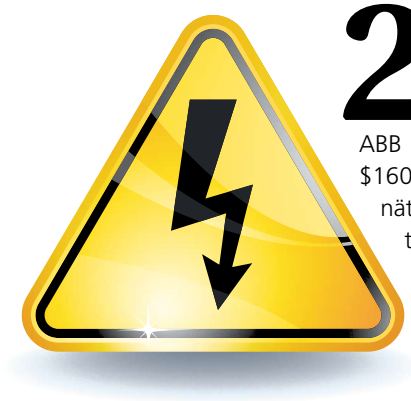
Lead-acid batteries are able to deliver the very large currents needed to start a car engine because of the exceptionally high electrical conductivity of the battery anode material, lead dioxide. A team of researchers have explained for the first time the fundamental reason for the high conductivity of lead dioxide, reports *ScienceDaily.*

"The unique ability of lead acid batteries to deliver surge currents in excess of 100 amps to turn over a starter motor in an automobile depends critically on the fact that the lead dioxide, which stores the chemical energy in the battery anode, has a very high electrical conductivity, thus allowing large current to be drawn on demand," said Professor Russ Egdell of Oxford University's Department of Chemistry, an author of the paper. "However the origin of conductivity in lead oxide has remained a matter of controversy. Other oxides with the same structure, such as titanium dioxide, are electrical insulators."

Through a combination of computational chemistry and neutron diffraction, the team demonstrated lead dioxide is intrinsically an insulator with a small electronic band gap, but invariably becomes electron rich due to the loss of oxygen from the lattice, causing the material to be transformed from an insulator into a metallic conductor.

Automation by the Numbers

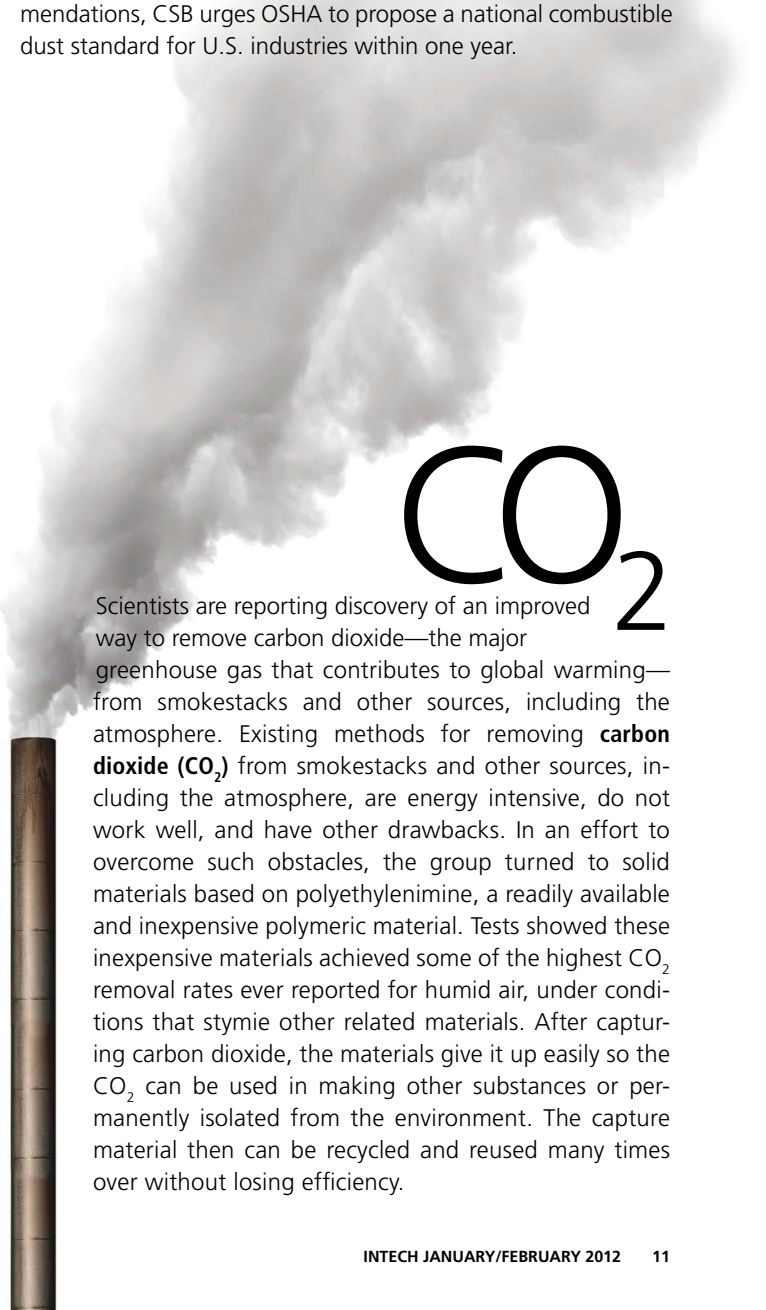
27 Volkswagen is going after the heart of the hybrid market in the U.S. with the new Jetta Hybrid that will complement the existing Jetta TDI. The Jetta Hybrid makes uses of a 1.4-liter turbocharged four-cylinder engine that generates 150hp and 184 lb-ft of torque. The gasoline engine is paired with a **27 kWh electric motor**, lithium-ion battery pack, and a seven-speed dual-clutch transmission. Total system output is a respectable 170hp. The Jetta Hybrid is capable of climbing to 60mph in less than nine seconds, can travel up to 1.2 miles on battery power alone, and can accelerate up to 44mph on battery power. Estimated combined fuel economy for the Jetta Hybrid is 45mpg.



200 ABB won an order worth around $160 million from Svenska Kraftnät, the national grid operator, to provide a new high-voltage underground cable system for the South-West Link power transmission project in southern Sweden. When completed in 2014, this will be the longest and most powerful underground cable link in the world. The main objective of the new transmission system is to enhance capacity and strengthen the reliability of the national power grid. ABB's underground high-voltage direct current cable system will have the capacity to transport 2 x 660 megawatts of electric power at a voltage level of 300 kilovolts across a distance of about **200 kilometers** between Barkaryd and Hurva in southern Sweden.

—Courtesy of Automation.com

5 Three industrial combustible dust accidents that killed **five workers** and seriously injured three over **five months** last year at a Tennessee manufacturer were entirely preventable and underscore the need for national dust regulations, concludes an investigative report released in January by the Chemical Safety & Hazard Investigation Board (CSB). The facility manufactures a fine (45–150 µm) iron powder used to make parts for the auto industry. It employs 180 workers and is owned by GKN, a U.K. engineering firm. The board found combustible dust piled up to four inches deep at the factory in an environment that used hydrogen and found the facility even flared the explosive gas inside the plant. Among its recommendations, CSB urges OSHA to propose a national combustible dust standard for U.S. industries within one year.



CO_2 Scientists are reporting discovery of an improved way to remove carbon dioxide—the major greenhouse gas that contributes to global warming—from smokestacks and other sources, including the atmosphere. Existing methods for removing **carbon dioxide (CO_2)** from smokestacks and other sources, including the atmosphere, are energy intensive, do not work well, and have other drawbacks. In an effort to overcome such obstacles, the group turned to solid materials based on polyethylenimine, a readily available and inexpensive polymeric material. Tests showed these inexpensive materials achieved some of the highest CO_2 removal rates ever reported for humid air, under conditions that stymie other related materials. After capturing carbon dioxide, the materials give it up easily so the CO_2 can be used in making other substances or permanently isolated from the environment. The capture material then can be recycled and reused many times over without losing efficiency.

Industrial energy conservation:
Where does the reasoning begin?

Exploit "low hanging fruit" energy conservation opportunities

By Bill Lydon, Chief Editor, *InTech*

Industrial energy conservation, "Where does the reasoning begin?", as one of my engineering professor's would say. Most industrial plants today have "low hanging fruit" opportunities that can readily be identified and will deliver quick payback with small investments. Industrial energy conservation has become a big topic with a wide range of products, technologies, and services being promoted to save energy in industrial and process plants that can be overwhelming. Many technologies, including sophisticated optimization to increase the energy efficiency of plants and processes, can be complex and expensive. Another component of an ongoing energy conservation program is sub-metering energy use, energy dashboards, and benchmarking that are useful tools and are being recommended as a first step by many automation suppliers. The majority of first step energy conservation measures only require common sense and basic engineering knowledge. Many plants are better served by first pursuing a basic energy conservation program to identify actions that save energy quickly.

As with any other project, you need to develop an understanding of the opportunities and challenges and then develop a plan. Starting with a basic program will put you and your company on the path to increasing energy efficiency and getting early results that build credibility with management to do more in the future. The information and basic steps described here should help you get started on the path to saving energy.

Economics

Energy costs in most industries are likely the most uncontrollable raw material cost for manufacturing or at a minimum in the top three raw material inputs that directly impact production profits. The U.S. Energy Information Administration report, International Energy Outlook 2011, predicts world energy use to increase 117% from 2008 to 2035. Energy has not typically been on the production bill of materials, but this is a growing trend. Savings generated from energy conservation drops directly to the bottom line increasing profits. The other economic impact is environmental, which more companies are considering important for social reasons, and in a growing number of countries, there is a surcharge for carbon dioxide emissions. A simple example illustrates the impact: It takes approximately 394 pounds of coal to keep a single 100 watt incandescent light bulb burning for 12 hours each day for a year. Burning the coal to produce this energy creates 936 pounds of acid rain causing 1,000 pounds of carbon dioxide and 7.8 pounds of sulfur dioxide. In addition, 90% of the energy consumed by the incandescent bulb is given off as heat rather than light.

These are the steps to start an energy conservation program and identify "low hanging fruit" opportunities.

FAST FORWARD

- Take advantage of "low hanging fruit" opportunities for energy conservation.
- Small energy conservation projects gain future management support.
- The U.S. Department of Energy notes as much as 1.6 to 3.2 quadrillion Btu could be saved by improving the efficiency and reducing energy losses in industrial systems.

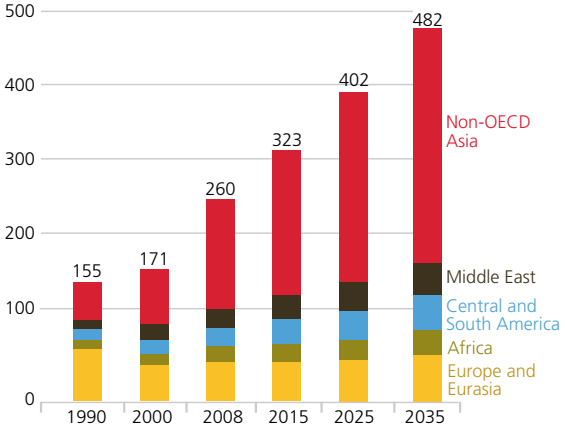
Get management support

Management support is an essential ingredient of the action plan to allow you to be proactive in going after opportunities to identify and make improvements. At this stage, the goal is to get enough management buy in to pursue basic energy-saving measures to have successes that prove the value of energy saving investments. This will illustrate the potential cost and productivity advantages of energy projects and build credibility with management to pursue more aggressive energy efficiency program later. Starting with a simple profile of overall energy use for your facility provides a basis to interest management, illustrates the size of the opportunity, and the baseline from which to measure your overall progress. Get the energy bills for electricity, natural gas, and fuel oil for the last year, and determine your total annual and monthly energy costs by fuel type. The U.S. Department of Energy notes as much as 1.6 to 3.2 quadrillion Btu could be saved by improving the efficiency and reducing energy losses in industrial systems (10-20% reduction in energy use). Energy conservation investments should be treated as another way to improve profits.

Form an energy team

Energy teams in manufacturing facilities identify energy-saving op-

Non-OECD energy consumption, 1990-2035 (quadrillion Btu)



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portunities, develop an energy plan, and implement cost-saving measures. Energy teams should include members from plant and process engineering, maintenance engineering, procurement, and production since energy systems are part of the fabric of a plant.

Energy conservation basics

Orienting the team to thinking about the sources of energy and fundamental ways to save energy is important before doing a plant walk through to identify energy conservation opportunities.

Energy sources used by plants are sometimes referred to as W.A.G.E.S., and this is a convenient way to remember major energy categories namely, water, air, gas (Natural Gas, other gases or fuels), electric, and steam. Throughout a plant, energy sources are transported and used in the production process. The main categories of basic energy conservation are eliminating losses, matching supply to demand, and increasing equipment efficiency.

Eliminating losses

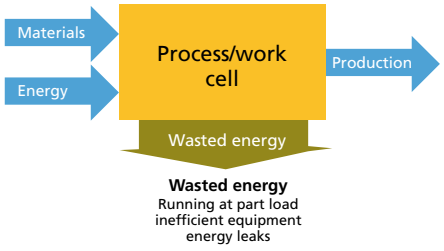
Eliminating energy losses is the most fundamental energy conservation strategy that is not glamorous but is generally low cost and high payback. Consider a simple water leak of one drop per second = 1 cup every 10 minutes, consumes over 3,200 gallons (12,000 liters) a year.

Load matching

Existing plants and processes generally have a number of opportunities to match the required output of equipment to the production need. When energy was lower cost, many machine and process designs and operating procedures were not optimized. For example, on a project in a wire mill at a steel plant, the set points for large oil heaters used in the process were set significantly above the required temperature. A simple lowering of setpoint and adding an automatic change to a standby setpoint when the process was not in run mode had significant savings.

Efficiency retrofit

Simple replacement of basic energy consuming devices with newer technology can save energy. Using more efficient light bulbs, sometimes requiring fixture changes, is a straightforward change to save energy. A bigger investment that can lower energy consumption and lower maintenance cost is replacing existing motors with NEMA Premium Efficient or EISA-compliant motors to decrease power consumption. In many cases, smaller-size motors can be used increasing the savings since the motors on existing equipment in many cases were larger than required. Some utility companies even offer incentives to customers who install new motors and gear drives. The government has been mandating minimum efficiency levels for motors manufactured in the U.S. since 1997 originally under the Energy Policy Act (EPAct), and now under the Energy Independence and Security Act (EISA). After EPAct was implemented, motor manufacturers began improving their efficiencies beyond the minimum requirements, so the National Electrical Manufacturers Association (NEMA) developed its own standard to identify motors that exceeded the mandated levels. Recognizing the industry's ability to meet an elevated set of standards, the EISA mandated all motors manufactured after 19 December 2010 must meet NEMA Premium Efficient standards. A complete list of NEMA Premium Efficient standards is available at www.nema.org.



Think of parts of processes and manufacturing as cells to focus on finding inefficiencies.

Walk through

Doing plant walk throughs to identify potential areas for improvement and the equipment that uses the most energy in your plant is a valuable use of time. In many plants, a minority of the equipment accounts for the majority of energy consumption. Things to look for include large pieces of equipment and equipment that runs most of the time or that runs periodically but use substantial energy. Tell people in the plant you are on a hunt for energy wasters and ask for ideas. Today it is easy to use a digital camera, video camera, tablet computer, and/or digital voice recorder to take comprehensive notes. Based on this information, a plan can be put together for energy saving measures.

Target area examples

These are examples of potential areas to save energy.

Compressed air

Compressed air is an essential energy resource within industries. The efficiency of a compressed air system starts at the compressor and stops at the point of use. Losses due to leakages within the pipework can cause extreme and completely unnecessary costs and reduce the efficiency drastically. Leakages are a constant consumer of compressed air 365 days a year. Over the years, compressed air systems often get extended with different materials being used, pipe diameters that are not optimal, and poor installation practices. Tracking down leaks is a detective job, and making all plant personnel aware of the issue and requesting they report suspected leaks will help. Monitoring air compressor operation during plant shut down time can provide insight into compressed air leakage issues.

Steam

According to the U.S. Department of Energy, more than 45% of all the fuel burned by U.S. manufacturers is consumed creating steam. Steam is used in many production processes and for building heat and electricity generation. Steam is not free; it costs a great deal of money to feed the boilers generating the steam. Steam is a very efficient way to transport heat energy, and it is easily moved in pressurized piping systems that can deliver that energy at manageable costs.

When steam gets to its point of use and gives up its latent heat to the envi-

ronment or to a process, it condenses into water, which must be returned to the boiler for reconversion to steam. Faulty steam traps are a large energy waster. Steam traps are valves designed to remove condensate as well as air from the piping system. Steam traps can fail open or closed creating problems. There are a number of technologies available to detect faulty steam traps including thermal and ultrasonic devices. Devices are also available to continually monitor steam traps and report status over industrial networks or more recently over wireless communications making them easy to install.

Thermal imager devices can also be used while steam systems are in operation to scan steam transmission lines for blockages, identifying closed valves, leaky steam pipes, blocked heat exchangers, and various boiler issues.

Consider creating a regular inspection route that includes all key steam-system components in your facility that are inspected at least annually.

Pumps

Pumps are used widely in industry to provide cooling and lubrication services, to transfer fluids for processing, and to provide the motive force in hydraulic systems. Since they serve such diverse needs, pumps range in size from fractions of a horsepower to several thousand horsepower. Common maintenance items to improve efficiency include bearing lubrication and replacement, mechanical seal replacement, and packing tightening and replacement.

Conservative engineering practices often result in the specification, purchase, and installation of pumps that exceed process requirements. Engineers often decide to include a margin of safety in sizing pumps to compensate for uncertainties in the design process. Anticipated expansions in system capacity and potential fouling effects add to the tendency to specify pumps that are "one size up" from those that meet system requirements. The cost of oversizing pumps extends beyond energy bills. Excess fluid power must be dissipated by a valve, a pressure-regulating device, or the system piping itself, which increases system wear and maintenance costs. There are five common indications that a pump is oversized: excessive flow noise, highly throttled flow control valves, heavy use of bypass lines, frequent replacements of bearings and seals, and intermittent pump operation.

Pumps that experience highly variable demand conditions are often good candidates for a variable frequency drive (VFD) to regulate motor speed to match the pump's output to required levels. The principal advantage of VFDs is better matching between the fluid energy that the system requires and the energy that the pump delivers to the system. As system demand changes, the VFD adjusts the pump speed to meet this demand, reducing the energy lost to throttling or bypassing excess flow. The resulting energy and maintenance cost savings often justify the investment in the VFD. However, VFDs are not practical for all applications, such as systems that operate high static head pressures and those that operate for extended periods under low-flow conditions.

Insulation

Anything that has insulation that is aging is a source of energy loss; prime examples are chilled water and steam pipes. Un-insulated steam distribution and condensate return lines are a constant source of wasted energy. Insulation frequently becomes damaged or is removed and never replaced during steam system repair. Water damage commonly creates insulation damage caused by leaking valves, external pipe leaks, tube leaks, or leaks from adjacent equipment. Any surface over 120°F should be insulated, including boiler surfaces, steam and condensate return piping, and fittings.

Develop a strategy

The final step is to create a strategy for sustaining plant-wide efforts to improve and maintain the efficiency of your energy systems. Keep staff motivated to achieve savings at your plant through monthly or bi-monthly meetings of the energy team, tracking and reporting on your energy and cost savings, and periodic reassessments of equipment and opportunities. Once areas for saving energy in one plant are identified, they can generally be replicated in other plants.

ABOUT THE AUTHOR

Bill Lydon (blydon@isa.org), *InTech*'s chief editor, has more than 25 years of experience in building automation, energy conservation, and industrial automation, including product design, application engineering, and project management. Energy conservation project experience includes commercial buildings, restaurant energy management, and industrial processes energy conservation.

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RESOURCES

Variable Speed Drives: Principles and Applications for Energy Cost Savings, 3rd Edition
www.isa.org/link/Variable_bk

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http://www1.eere.energy.gov/industry/rd/footprints.html

Energy-Related Best Practices: A Sourcebook for the Chemical Industry
http://www.ciras.iastate.edu/publications/EnergyBP-ChemicalIndustry/

Processing heavier crudes to meet future needs

Improved modeling improves designs

By Joseph McMullen, Brittany Doyle, and David Bluck

Countless factors will drive the future of the global refining industry, including the state of the worldwide economy; availability, accessibility, and quality of raw material; and legislation surrounding industry practices. For this reason, it is difficult to determine what exactly is in store for this important industry. However, assessing these contributing factors can help predict the direction the market might take.

Crude oil is a non-renewable resource. Once it is used, it cannot be regenerated, and its overall supply is continuously decreasing. In particular, the supply of higher-quality sweet, light oils, which, with their low sulfur content and low density, have historically been the easiest to extract, refine, and process, is rapidly diminishing. This has created the need for new technical resources and capabilities that help obtain and refine lower quality, heavier sour oils that are more difficult to access.

For that reason, the industry's attention has shifted from downstream processing to upstream exploration and extraction. Few methods are currently available for extracting heavy oils and bituminous oil sands from deep within the earth. Yet, this will become a crucial practice as the supply of accessible light, sweet oils continues to wane. Having the knowledge to obtain these heavy oils or oil sands must be coupled with the technology to refine this lower-quality feedstock.

Heavy crude access, processing

Crude oil from the Western Canadian Sedimentary Basin will soon be a viable crude source. Transporting that crude through southbound pipelines will result in new feedstock for refineries in the U.S. But that means U.S. refineries will need new capabilities to refine these crudes. Currently, only a few U.S. refineries, primarily on the western coast, have that capability. Because of their proximity to Canada and their processing ability, these refineries could see an increase in throughput compared to their less flexible counterparts.

However, pulling the oil out of the ground is still a problem. Since this crude is more viscous, it must be loosened before it can flow upward. Enter steam assisted gravity drainage (SAGD) technology. In the SAGD process, steam is injected into the reservoir to heat the heavier, more viscous crude, allowing it to be removed through a second pipe. Historically, this has been an expensive process, but recent advances in technology and the increased price of crude means SAGD has become a real, economically viable alternative to oil production.

Modeling technologies have been a great help in implementing SAGD. Coupling a pipeline simulator with a reservoir simulator can accurately model the entire SAGD process, leading to enhanced efficiency in the design and operation of the process, but all of that hinges on the ability to accurately simulate heavier crudes, which many experts believe is the future of the industry. Unfortunately, however, the majority of commercial simulators utilize methods developed for conventional light, sweet crudes.

Invensys Operations Management recognized this back in 2007 when it started a heavy-oils consortium that engages customers to develop new simulation methodologies that accurately model heavier crudes. The empirical data from the consortium is fundamental to building accurate models for heavy oil production. Consortium members include Shell, ConocoPhillips, Suncor, Syncrude, Petrobras, Chevron, TOTAL, Petrocanada, PDVSA, KBR, Toyo Engineering, Fluor, BP, StatOil, and ENI. Processing these heavier oils can tax equipment that is not designed appropriately. Therefore, it is important the crude oil is characterized correctly from the beginning of the design process. To address this, Invensys has developed heavy oil methods, using its SimSci-Esscor technologies that are able to accurately characterize heavier oils.

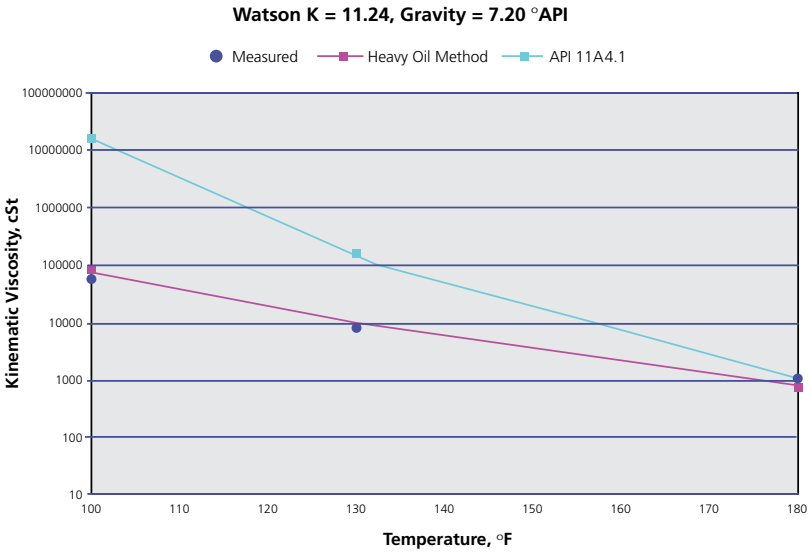
FAST FORWARD

- Crude oil is a non-renewable decreasing resource, making it important to better exploit heavy oils or oil sands.
- Coupling a pipeline simulator with a reservoir simulator can accurately model the entire steam-assisted gravity drainage (SAGD) process.
- User partnerships are effective to create improved petroleum characterization procedures for heavy oils.

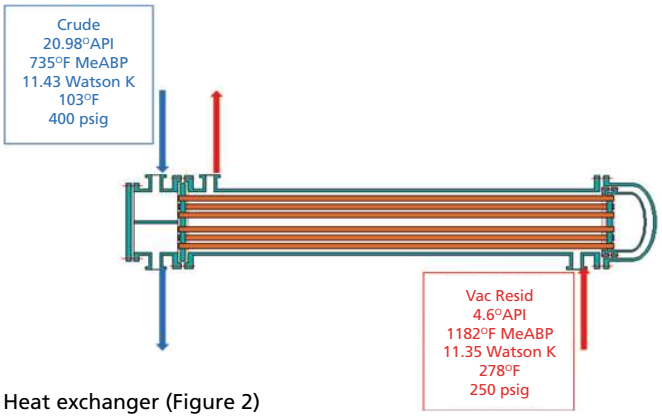
Characterization, modeling of heavy oils

In November 2007, through customer partnerships, processing data was used to create a new petroleum characterization procedure for heavy oils. This heavy-oils characterization method is intended to extrapolate critical properties and molecular weight of petroleum components with normal boiling points beyond 1000 degrees Kelvin.

Since it was implemented, the correlation has twice been modified and improved as more data became available from consortium members. Using kinematic velocity, Figure 1 shows how the new correlation compared to measured data for heavy crude.



Measured kinematic viscosity data for a whole heavy crude (Figure 1)



Heat exchanger (Figure 2)

To see how the increased accuracy would affect the design of process equipment, the example above illustrates the affects of kinematic viscosity on the design of a heat exchanger (Figure 2). The streams entering the heat exchanger are both heavy oils. Therefore, a comparison could be made on the predicted kinematic viscosity and how that would affect exchanger sizing during the design phase.

Temperature °F	Crude Kinematic Viscosity, cSt		
	Measured	AP 11A4.2	Heavy oil
60	49.75	111.6	70.65
80	27.95	58.77	39.69
100	17.52	34.26	24.38
	Vac Resid Kinematic Viscosity, cSt		
210	6787	454.0	4784
250	1261	147.5	858.5
300	274.6	50.83	176.8

Measured, predicted kinematic viscosities (Figure 3)

Viscosity(L) method ➤	Measured values	API procedure 11A4.2 (2011)	Heavy oil prediction
Duty (10^6 BTU/hr)	3.654	8.976	5.432
LMTD (°F)	165	150	159
U (BTU/hr/ft²/°F)	4.657	12.62	7.144
Shell side			
T_{out}(°F)	265	245	258
ΔP (psi)	58	8	27
Tube side			
T_{out}(°F)	110	119	113
ΔP (psi)	9	13	11

Calculated exchanger design (Figure 4)

In Figure 3, it is easy to see the differences between the kinematic viscosity values predicted by API 11A4.2 and the heavy oil correlation. For the crude stream, the API 11A4.2 over predicts the viscosity, wherein the kinematic viscosity is under predicted for the Vac Resid stream.

In both cases, the heavy-oil correlation-predicted values are much closer to the measured values than the API A4.2 method. This illustration shows the heavy-oils method much more accurately predicts the kinematic viscosity. However, the important factor is how well the heat exchanger is designed, as well as how the equipment design is affected by the kinematic viscosity.

Figure 4 depicts how kinematic viscosity affects the design of the heat exchanger. Notice if the API Procedure 11 A4.2 is used, then the duty would be oversized by more than 200%. The heavy-oil method also over predicts the duty, but by a much smaller margin. The heat transfer value is over predicted by almost 300% with API 11A4.2, while the heavy-oil method over predicts by less than 200%. These calculations could lead the design to be unnecessarily large, which would increase capital and operating costs. The ramifications of the under-predicted pressure drop could lead to not having the proper pumps in place to drive the fluid through the exchanger. These flaws in the design of this exchanger are cause for concern and support the need for accurate heavy-oil modeling.

Modeling additional properties

Invensys designers have also developed a methodology for modeling heavy oils that can be detailed for liquid viscosity and liquid thermal conductivity. This helps to more accurately predict the properties of heavy oils, which leads to enhanced accuracy in process design and operation.

Additionally, because mercury is a common pollutant in heavy crude oil that is increasingly regulated, Invensys has also introduced an updated methodology for predicting mercury solubility in hydrocarbons. The new methodology enables proper accounting for mercury within raw materials, products, and waste, as well as mercury mitigation.

As the Invensys-sponsored heavy-oils consortium continues to provide more data and valuable guidance, more accurate thermodynamic methods will be developed that anticipate growing and changing industry needs to help oil processors optimize their operations in real time. This will drive the continued development of software modeling tools and provide value to engineers by helping them address design and operational problems now and in the future.

ABOUT THE AUTHORS

Joe McMullen is the SimSci-Esscor product marketing manager for Invensys Operations Management. He began his Invensys career as a senior technical support specialist before becoming product manager for the company's SimSci-Esscor flagship PRO/II process simulation software. In his current role, he manages all of the company's steady-state simulation software products, responsible for developing and expanding the company's capabilities in simulation, advanced control, optimization and training systems. **Brittany Doyle** is an intern at Invensys Operations Management and a student of Villanova University. This article is based on work by **David Bluck**, chief technologist, Invensys Operations Management.

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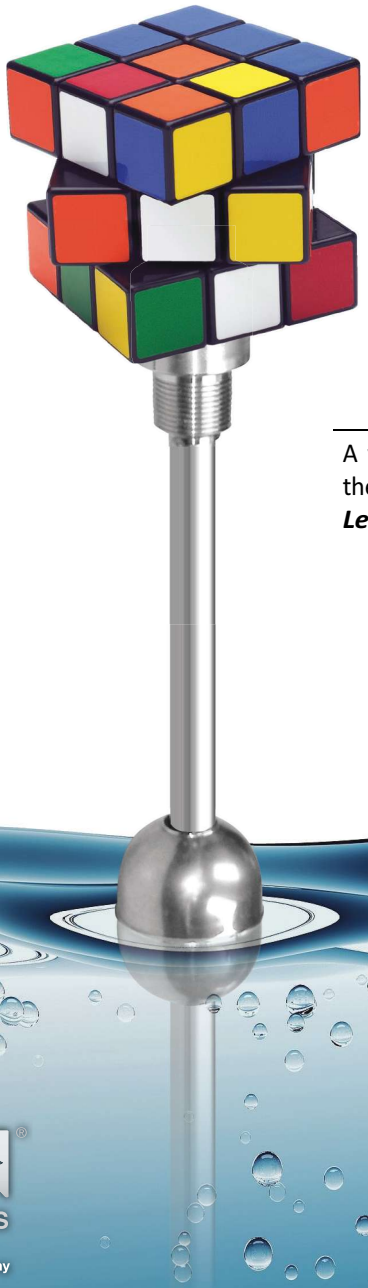

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Small multi-axis industrial robots add new twist to lean manufacturing

Small robots offer ultimate in repeatability, flexibility

By Charlie Miller

Lean manufacturing is a term that has many interpretations. To some, it primarily means low cost. To others, its core is to realize the highest productivity possible while virtually eliminating waste. Other definitions focus on minimizing human capital and optimizing process motion and flow. Whatever the interpretation, in general, it is *to do more with less*.

While all of these concepts are clearly related, according to popular sources, the macro view of lean manufacturing has evolved into "the creation and maintenance of a production system, which runs repetitively, day after day, week after week, in a manner identical to the previous time period."

In recent years, industrial robots have helped add a flexibility component to lean manufacturing by allowing multiple parts to be run on a single line, thus better leveraging the investment on capital equipment. While robots offer the ultimate in repeatability, they do not need to perform the same motion to do so. The strength of robots is their inherent ability to be changed, adding a valuable dimension of flexibility to the production process. Rather than running repetitively day after day, they provide precisely repeated motion within a specific production cycle, with the ability to achieve the precise repetition of a completely different motion for the next production cycle, all with what amounts to the simple flip of a preprogrammed switch.

As robots continue to expand their application base from their automotive and heavy industrial roots, smaller, more lightweight robots are being developed, adding a new component to the translation of lean manufacturing. Larger robots are built for stability, strength, and reach, being able to lift and move heavy objects longer distances. Smaller robots, by comparison, are designed for tasks with a lighter payload where a compact work envelope is required.

If the parts to be handled are less than 5kg and the distance travelled is less than 500mm,

FAST FORWARD

- The development of small multi-axis robots is reflective of the evolution of small robots making them useful in more applications.
- Smaller robots offer the ultimate in repeatability and flexibility for a range of manufacturing tasks.
- Two recent installations using small robots illustrate how small footprint functionality increases productivity.

then some of the newer, compact 6-axis robots available are viable options. Smaller robots are ideal for assembly, machine tending, pick-and-place, dispensing, and packaging applications. Often referred to as tabletop robots, they can be mounted from a variety of angles, from a ceiling, wall, or shelf, allowing for the design of the most efficient work flow possible.

The development of small multi-axis robots is reflective of the evolution of small robots and the considerations that have led to their emergence. For example, the recently introduced IRB 120 is ABB's smallest ever 6-axis robot, weighing just 25kg (55 lbs), with a standard payload of 3kg (6.6 lbs) and a reach of 580mm. When the design work first began on the IRB 120, it was intended for assembly work in the electronics industry. The idea was to make an affordable robot suited to low-cost countries where electronics are typically manufactured. But they soon caught on, domestically and abroad, as other industries found these robots improve productivity and are easily integrated into new and existing production lines. The installed base has extended to a variety of industries including pharmaceutical, packaging, food and beverage, automotive, and solar photovoltaic manufacturing. One of the key features of this new class of smaller, compact robots is they are able to work very close to other machinery in a production line. A typical base for these robots is 18 centimeters by 18 centimeters, the size of half a piece of A4 paper.

This class of robots typically weighs just 25 kilograms and has a very compact turning radius, enabled by the robots symmetric architecture, without offset on the second axis. This ensures the robot can be mounted close to other equipment, and the slim wrist enables the arm to reach closer to its application.

These compact robots also offer an advantage when mounting the robot upside-down, as it can be installed at a relatively low height, once again saving space. At the same time, these robots do not sacrifice reach. Typical "stroke" measures 411 millimeters, which is long compared to its total reach of 580 millimeters.

The size of controllers for this new class of robots has been reduced, as well, by as much as 80% compared to conventional robot controllers. These small robots are also becoming available for Clean Room ISO 5 applications.

Two recent installations using small robots demonstrates how the small footprint functionality is easy to integrate into existing lines while providing the same functionality as larger, more traditional industrial robots.

L'Oréal cosmetics

The international cosmetics industry is a competitive, fast-paced business. When demand from hair salons across Europe skyrocketed in 2009 for L'Oréal's INOA (the world's first ammonia-free permanent hair color), the French cosmetics giant made plans to ramp up production.

L'Oréal Canada needed to set up a packaging line in Montreal for hair coloring products quickly. The highlight of the solution was a brand-new, small robot.

ABB IRB 120 robot places tubes of L'Oréal INOA hair coloring dye into boxes.

At the time, L'Oréal Canada's flagship plant in Montreal had two production cells that churned out some 150 million units a year of hair-coloring liquids and creams. The facility was asked to begin manufacturing large quantities of INOA in only four months.

"It was a huge challenge," recalled Guy Fafard, the plant's technical supervisor. "When we discussed it with our production manager, he said

there was no way we could design and install a new production line in such a short period of time. He said it simply couldn't be done. But we had to find a way to make it happen."

Fafard turned to PharmaCos Machinery, a local leader for turnkey solutions in pharmaceutical and cosmetics packaging equipment that often does custom, needed-it-yesterday production projects for L'Oréal Canada. Within days, the company's technological development director, Sylvain Gauthier, was walking the floor of the busy plant with Fafard to get a firsthand feel for the project.

"It was a complicated mandate," said Gauthier, who worked as a technician at the L'Oréal Canada plant for 10 years before joining PharmaCos a decade ago. In addition to the tight deadline and the use of an explosion-proof tube filler (because INOA uses small amounts of alcohol in place of ammonia), the new line needed to be able to take tubes, put them in trays, and load them (plus an instruction sheet) into a ready-to-ship package.

According to Gauthier, such a two-step, two-micro-stop cartoner process would normally be done manually. However, it was critical the new line always keep moving, because a stop would cause the pressurized fill to overfill the first tube (due to the positive pressure in the reservoir). He also had to respect the plant's production philosophy of having only a single operator for small lines such as the INOA project. "My only option,"Gauthier recalled, "was to design and build a new conveyor based on a 29mm center-to-center tray and a small robot."

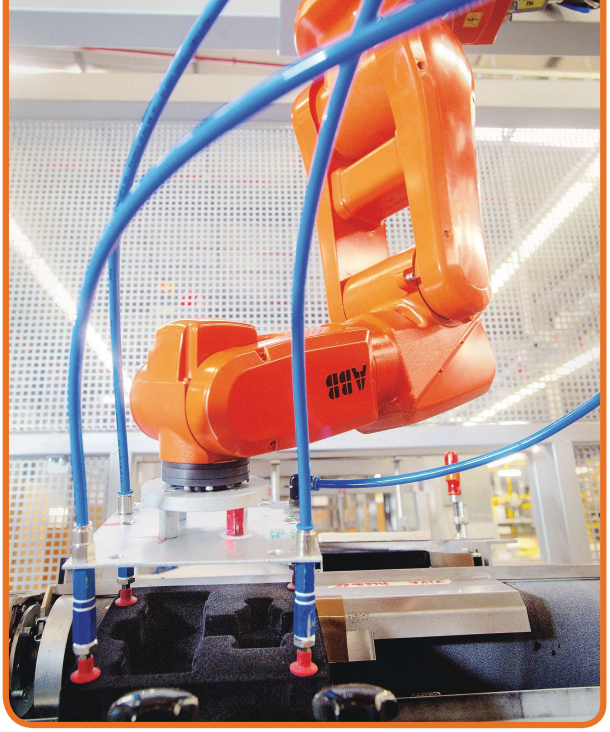
While talking to ABB's Canadian office, Gauthier learned ABB was in the process of bringing its smallest ever six-axis robot to market. Such a small robot would fit the limited space requirements of the workspace, be easy to set-up, begin production quickly, and be easily accessible for frequent maintenance. All while providing the precision necessary to deftly handle the product, and the future flexibility to be easily reprogrammed for different tube sizes and configurations as consumer demands change.

After a 3-D model with ABB's RobotStudio design simulation software, it took Gauthier only two months to build, assemble, and test the new line. The solution integrates a Kalix cartoner system, an IRB 120 robot from ABB, and an Allen-Bradley programmable logic controller (PLC) to keep count and control input and output. "Once we got it going, it worked like a charm," said Gauthier.

BDMO custom packaging manufacturer

Vivabox gift packages are distinctive, and in Belgium, consumers know they contain quality products. The packages have become so successful—they number about half a million per year—they have become a brand in their own right.

Vivabox is one of packaging producer BDMO's largest product lines at its factory in Meulebeke, a town in Flanders, Belgium. The manufacturing process of the actual box and its lid involves several stages, all of which are automated on various product lines. For example, the carton has to be cut,



An invert mounted ABB IRB 120 runs 16 hours a day, inserting 1,200 packaging trays an hour.

scored, folded, and taped. After that, the relevant printed cover is applied.

Until recently, the last part of the process was not automated. It involved inserting a thermoplastic tray, and different trays are used to hold different gifts. For example, one may be used to hold four miniature bottles of malt whisky, while another might hold coffee sachets, two cups, and saucers.

These thermoplastic trays were inserted by hand because they are relatively thin—they flex, and the fit has to be tight. Up to that point of the process, automation produces 1,000 boxes every hour. That meant up to seven people were needed to keep up with the flow of boxes, one every three seconds.

Could the insertion of the tray also be automated? That was the question BDMO put to Viscon, a local systems integrator, after Daniel Callewaert, BDMO's maintenance manager, saw a roadside video wall that promoted their robotics and transport automation expertise.

After several rounds of consultation and due diligence, Viscon's proposed a "pick-and-place" system that could handle 1,200 trays an hour. The tight fit problem was resolved by using the 6-axis functionality of ABB's IRB 120 robot; it inserts the tray at an angle before pushing it firmly down to the base and onto spots of glue.

The resulting solution now runs 16 hours a day in two shifts, and the personnel head count has gone down from seven to three. Needless to say, the cost savings have been significant. The solution was fine-tuned after the initial trial, and this boosted the placement rate to 1,400 trays, thereby adding additional reserve capacity.

"The earlier production speed of 1,000 Vivaboxes was determined by the manual insertion process," said Pieter Debucquoy, BDMO's maintenance coordinator. "The robotic solution gave us an immediate 20% boost, and after the fine tuning we have the possibility to add an additional 15%." Ironically, the preceding part of the process has become the new bottleneck.

BDMO produces more than 10 million packages a year. With the flexibility of robots to easily adjust to a variety of packaging shapes, sizes, and construction geometries, looking ahead, BDMO is likely to deploy additional smaller robots to automate other packaging production processes.

ABOUT THE AUTHOR

Charlie Miller, vice president of sales for ABB Robotics, has more than 25 years of experience in the industrial robotics industry. Miller started in the Flakt Division of ASEA in 1985 (Flakt was later rolled into ABB in the 1988 merger of ASEA and Brown Boveri.), and he has been in numerous positions including project engineering, project management, proposal engineering, proposal management, business unit manager, product management, and sales management. For more information, please contact sales.info@us.abb.com.

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Simulation improves operator training

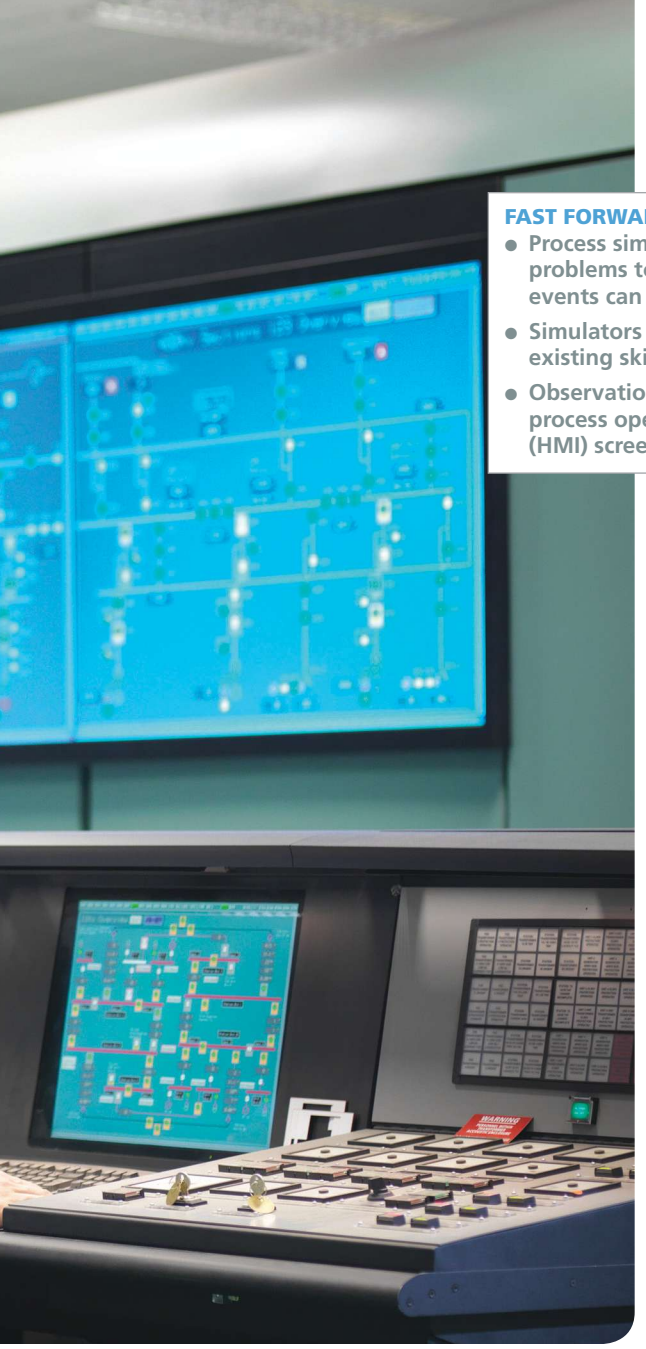
PC-based simulation cuts operator training costs while yielding superior results as compared to other training methods

By Platt Beltz

Humans are visual creatures who learn more by seeing how something works than by just reading about it. Further learning improvements result by actually performing the tasks at hand. That is why process plant operator training through simulation is practiced throughout the industry, albeit with varying degrees of success.

Simulation training is vital for preventing incidents and accidents. It also improves process control, resulting in higher throughput and quality with less downtime. Maintenance is reduced because equipment is operated closer to specifications.

In addition to immediate costs, operator errors and subsequent incidents can result in fines or even jail time for plant managers in some industries. These occurrences can be minimized or eliminated with the right training plan and equipment, of which off-line process simulation is a key component.



FAST FORWARD

- Process simulation enables actual unit operation problems to be depicted, and common and unique events can be re-created.
- Simulators can be sped up or slowed down to build on existing skills and boost operator confidence.
- Observation of operator actions can be used to improve process operations and Human Machine Interface (HMI) screen designs.

Why simulate?

To understand the value of operator training through simulation, rewind a couple of years. Remember your first day in the plant? There was a multitude of pipes, tanks, smells, and equipment you had not seen before—at least not in these particular configurations. You went to the control room where you were promptly overwhelmed by the instruments covering the wall, for those old enough to remember panel boards, or by computer monitors showing myriad colored objects on their screens.

Your new plant personnel feel the same way; even more so with fewer mentors available now and more pressure to learn quickly combined with a lower tolerance for mistakes. Simulation training helps plant managers meet these demands and helps operators come up to speed quickly.

When process simulators were introduced in the early days of the Distributed Control System (DCS), they required extensive software engineering just to get their screens to look like the ones operators were using. Much more time was required to simulate the process itself, in addition to a very high level of process knowledge.

Simulator programming typically was done in UNIX, requiring personnel with special skill sets to program and maintain the simulator. If this sounds expensive, it was. Typically nuclear power plants and refineries, where simulation capability was absolutely critical to prevent incidents, were the only industries that could justify these costs.

The arrival of PC-based simulation has made simulation training affordable to a multitude of industries, and this level of such training can be adjusted to meet the budget and the level of simulation requirements.

Levels of simulation

Low fidelity, medium fidelity, and high fidelity define the three levels of PC-based simulation. These terms loosely describe how close the simulated plant's process and equipment responses are to the actual plant.

Start with the basic process simulator that is generally part of the engineering configuration software supplied with the control system, particularly with a higher-end process plant DCS. Basic process simulators provide software loop tie backs in which the output of a loop is taken back into the input through software in a virtual environment. This creates basic loop responses that give operators a fundamental feel for loop control, screen navigation, and responses. Simulating more sophisticated loops is not feasible with this type of software.

The next level of simulation uses two PCs—one running the control software program, and the second supplying process simulation responses. The two PCs typically communicate via Ethernet. Sophisticated and realistic process

dynamics now become an integral part of the simulation. Sizing of vessels, stroking times of a valve, and dynamics of the process can be entered and adjusted. The properties of the process unit in the simulator PC can be adapted to include process noise, making the simulation more realistic.

This type of simulation is not meant to replicate exact plant processes, but it can be modified on a tag-by-tag basis to yield required response levels. It can also be expanded to cover the entire plant.

The top level of the process simulation hierarchy is the high fidelity simulator. It can precisely replicate the process dynamics for every piece of equipment in the plant. If operator training simulation must closely mimic the actions of the process unit or plant, this is the route to take. Several types of industries require this level of simulation, and many others could benefit**.**

A well-designed, implemented, and operated process simulation training program will provide many benefits.

1. **Provides the least expensive operator training method**
   The introduction of PC-based simulation into the process control industry made simulation training affordable. The PC hardware is inexpensive, and graphical programming methods created for the Windows operating system now enable plant personnel, instead of IT experts, to program and configure the simulator. Operators can now be trained on-site in smaller time blocks, instead of being sent to training classes at distant locations.

2. **Improves quality of operator response and subsequent actions**
   The process simulator can create scenarios that depict actual unit operation problems. Common and unique events can be re-created so the operators' responses can be seen and recorded. Senior operators' responses can be used to establish best practices for less experienced personnel. Once best practices are established, the training system can be used to measure improvement.

3. **Improves operator response time to process upsets and incidents**
   A process simulator can be set up to quickly change process operating conditions. Snapshots of the process running in specific conditions can be taken for instruction. Perhaps one operator needs to practice changing product grades on a static state process, but another needs to work on unit startup. PC-based simulation allows easy implementation of these scenarios.

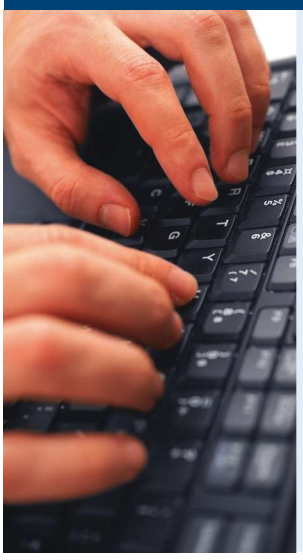
4. **Offers the fastest practical operator training method, particularly for inexperienced personnel**
   With fewer experienced operators available to train new operators, simulation training can provide invaluable instruction to fill the training gap. Simulators can be sped up or slowed down. For a process with a large amount of dead time, the simulator can be sped up to compensate for the delay. For training of inexperienced operators, actual process conditions can be slowed to build confidence and then gradually sped up.

5. **Helps meet regulatory requirements**
   In general, the higher the potential environmental and safety impact, the more oversight involved from government agencies, with corresponding increases in required training. Process simulation training provides the extensive instruction and detailed training records required to meet regulatory standards.

6. **Can be a mitigating factor if an incident does occur**
   In addition to direct costs, operator errors and subsequent incidences can result in fines or even jail time for the plant managers in some industries. If an accident does occur, simulation training programs and related records can be a mitigating factor to show the plant took precautions and performed due diligence.

7. **Often leads to process improvements including increased uptime, more throughput, and higher quality**
   Taking training a step further, observation of operator actions can be used to better the actual process control programs and the HMI screen designs. This can further enhance operator actions, reducing the possibility of incidents and improving general plant operation and output quality.

Benefits of simulation for operator training



1. Provides the least expensive operator, most life-like training method
2. Improves operator response time to process upsets and incidents
3. Improves quality of operator response and subsequent action
4. Offers the fastest practical operator training method, particularly for inexperienced personnel
5. Often leads to process improvement including increased uptime, more throughput and higher output quality
6. Creates better trained personnel to enable operations with a leaner staff
7. Can be used to meet regulatory requirements
8. As part of a comprehensive training program, can be a mitigating factor if an incident does occur

Implementation challenges

Although PC-based process simulation for operator training provides affordable, tangible benefits, there are challenges to implementation:

1. **A simulator that must mimic actual plant operations is more expensive.**

Operating training simulation is typically performed with two PCs, one to operate and process and the second to mimic actual plant operations.

All expenditures have an associated cost/benefit ratio, and operator training simulation is no different. Plant management must decide how closely the simulator needs to mimic the exact operation of the process because this is the primary cost driver. The closer the simulation to the actual response of the heat exchangers, reactors, process/product pressures and viscosities, the higher the cost, but the greater the potential benefit.

2. **Simulator programming must be integrated with existing systems and be kept up to date with changes.** Once the right level of operator training simulation is selected and implemented, a common point of failure is a lack of ownership or assigned responsibility. Although today's graphical software is much easier to use, simulation software must still be integrated with existing plant automation systems. Furthermore, every process undergoes changes, as do most simulation software packages, and someone has to be the champion and own and implement these changes.

3. **A regular training program must be instituted and followed, preferably with some type of certification.** Before simulation can be incorporated into operator training, the training program itself should be examined. Operator training should revolve around certification and benchmarking. Certification verifies specific skill sets have been met, and benchmarking creates best practices.

4. **Time and money must be allocated for on-going operator training.** The operator training simulation must have a champion who is supported by upper management. If it does not, it will fall out of use, and the last thing any plant wants is an investment gathering dust. Horror stories abound about plants spending large amounts of money on a process simulator that is not used anymore. That is not a failure of operator training simulation, but rather a failure of plant training and operating procedures.

From theory to practice

The military and the airline industry have been using simulators for decades. They understand the value of first experiencing situations in a virtual environment before being plunged into reality. The goal is to get the trainee as close to the real world as possible. This is accomplished by training individuals so if and when they experience a worst-case scenario in real life, they have already implemented the solution via simulation. This gears trainees for success.

Experienced operators are retiring at many process plants, and new personnel must become competent quickly. Fewer experienced operators mean fewer opportunities to spread plant and process knowledge than in the past. This can lead to unscheduled shutdowns, costing millions of dollars. Shutdowns can also bring fines, plus unwanted government and media attention.

To ameliorate this situation, an on-going operator training simulation program that challenges operators, new and experienced, should be considered. The results will speak for themselves as a process simulator allows competencies to be established in months, not years. Moreover, experienced operators can be trained in new processes, creating the ability to operate with a leaner staff. A smoother running process translates into a more profitable plant, yielding a quick payback on the simulation system investment.

No plant can risk training operators on actual equipment, but a good plant simulator will have the look and feel of the actual process, expediting the training of new and experienced operators without jeopardizing actual operations. The closer to the actual look and feel of the actual process, the more prepared operators will be when monitoring and controlling that process.

Often overlooked are the morale dividends created by investing in employee training. The time and expense involved with training on a process simulator is not lost on the plant employees as they know it is an investment in their future, as well as the future of the facility where they work.

ABOUT THE AUTHOR

Platt Beltz is training department manager at Yokogawa Corporation of America. Beltz has worked for 36 years in the process control field. He spent 13 years in field service for the Taylor Instrument Company as a field service technician and district service manager. He has been with Yokogawa for 23 years, starting in field service and then moving to training followed by stints in marketing and product support.

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Hackers can remotely power down critical automation

Uninterruptible power supplies and cybersecurity

By Michael A. Stout

The recent number of cyber-attacks and their level of sophistication have demonstrated the inadequate network security measures employed by many large corporations, government, and military agencies. Time is on the hackers' side. They only have to find one unsecure computer or device on a segment of a corporate or governmental network, and they can use any number of methods to eventually gain access to critical data. Should they not be able to find an unsecured computer, they simply have to send a cleaver e-mail containing a one-off designer backdoor virus that will evade many corporate level antivirus software and firewalls—and again they are in. WikiLeaks is a prime example of the amount of data an insider can obtain from a secured network and the damage that can result. In the ever-evolving hacker wars, many corporations and government agencies do not have a high enough level of protection for their most sensitive data. Recently, the hacker group "Anonymous," or "AntiSec", released a 7.4-GB file containing the e-mails and personal information obtained from the supposedly secure computers of 56 different law enforcement agencies.

Allowing the violation of a corporation's data is unacceptable, but what if the hacker gains access to a corporate Supervisory Control and Data Acquisition System (SCADA)? SCADA systems are the very heart of a corporation's automated manufacturing or process control. SCADA systems are installed to control and monitor the entire manufacturing process for automotive production, food processing, oil and gas refineries, pharmaceutical production, power production and distributions facilities, and others too numerous to mention. The Stuxnet virus attack is a good example. It is unknown who wrote the Stuxnet virus, but it is highly suspected to be a government's creation. At this point, it appears the virus has had only one intended target, a nuclear fuel processing facility located in Iran. Stuxnet took over the computer automated system

(SCADA) controlling gas centrifuges critical to the uranium enrichment process. The virus increased the centrifuges' speed to a point where they were destroyed. This occurred while the virus instructed computers controlling the centrifuges' speed to report their conditions as normal to engineers monitoring the process in the facility's control room. Stuxnet was released on the Internet and infected a large portion of the world's computers before finding its intended target. The handwriting is on the wall, and it is time for governments and corporations to make the security of their data and process control networks their number one priority. New technologies and security threats will continually be developed, making this priority a costly, ongoing battle demanding continuous risk assessment. The approach taken must be complete and all-encompassing as the network security chain is only as strong as its weakest link.

Every network connected device in a data or SCADA network is a potential backdoor into the network, or at a minimum a security risk. For instance, an uninterruptible power supply (UPS) powering a programmable logic controller (PLC) responsible for controlling a key controlled substance mixing process on a pharmaceutical production line may be subject to outside sabotage through its unsecured SNMP/HTTP network interface. A UPS connected to the company's Ethernet network for the purpose of remote monitoring and management could be compromised through the collusion between an inside employee and a hacker friend outside the company. The unsecured UPS IP address could allow the UPS to be shut down and restarted remotely to cover thefts of a small amount of the controlled substance.

The UPS units are essential should there be a power outage to the SCADA RTUs, as they must apply the brakes on large overhead crane motors to prevent floor workers from being killed or injured by falling loads. As the UPSs are the RTU's sole source of power, they are essential to assure the crane is powered until it is in a safe state. One can imagine the damage that could be caused by a disgruntled employee or hacker acquiring the Internet Protocol (IP) addresses to the UPS units and having the ability to turn their outputs on and off at will and without warning.

Once a hacker has gained access to a moderately secured network, they can easily determine the IP addresses of every device on the network. Using port scans, they then can determine if the device communicates through HTTP, SNMP, Telnet, MODBUS, etc. Once the communications protocol has been established, the hacker will first attempt to determine if the device has any further security. In the case of a device supporting HTTP protocol, if unsecured, it is a simple matter to use any web browser to communicate directly with the device, often by displaying a menu of options. The critical selection and proper configuration of a UPS SNMP/HTTP agent option is vital to network security, but often an afterthought. This prompts the question "What security features should a UPS SNMP/HTTP agent support?"

FAST FORWARD

- Allowing the violation of a corporation's data is unacceptable, but what if the hacker gains access to a corporate Supervisory Control and Data Acquisition System (SCADA)?
- Every network connected device in a data or SCADA network is a potential backdoor into the network, or at a minimum a security risk.
- An uninterruptible power supply (UPS) powering a programmable logic controller (PLC) may be subject to outside sabotage through its unsecured SNMP/HTTP network interface.

The world is running out of IP addresses under the old 32-bit IPv4 format, which has prompted the development of a new world standard IPv6 that supports 128-bit IP addresses. In addition to adding a virtually unlimited number of IP addresses, IPv6 has Internet Protocol Security (IPsec) built-in. When used, IPsec secures the IP communications across the network by authenticating and encrypting each IP data packet. IPsec uses a shared key to accomplish authentication. IPv6 support is essential in the selection process.

One can imagine the damage that could be caused by a disgruntled employee or hacker acquiring the Internet Protocol (IP) addresses to the UPS units and having the ability to turn their outputs on and off at will and without warning.

The SNMP/HTTP agent should be able to turn off unused communications ports in addition to the ability to reassign port numbers. A typical agent may support BootP/DHCP, Ping Echo, Telnet, SSH connection, HTTP, HTTPs, UDP, three SNMP versions, UPnP, and SMTP protocols. All of these protocols are assigned differing port numbers and can, if unsecured, identify themselves should a port scan be performed. Some of the ports could provide backdoor access to the agent and the associated UPS unit. It is a good practice to turn off unused communication protocol ports and to use communications protocols that have adequate security.

Telnet has been used for decades by network administrators to manage remote devices, but its security is very weak and, in the case of a hacker, non-existent. SSHv1 protocol offered a much more secure option as it has strong authentication

protections in addition to encrypting the communications across the network. SSHv2 was developed as it was determined that hackers could bypass security and execute code at the root level on UNIX-based systems. SSHv2 also supports the Triple Data Encryption Standard (3DES) in addition to AES, making it essential for a secure SCADA network.

Simple Network Management Protocol (SNMP) is used by larger companies to remotely monitor and manage most network devices like managed switches, printers, UPS units, file-servers, computers, modems, etc. SNMP provides a very robust means of monitoring any number of devices from one central workstation having network management software (NMS) installed such as SolarWinds or OpManager. Each device industry has developed a standard set of management instructions specific to the type of device referred to as a RFC Standard Management Information Base (MIB). The MIB for the applicable device is supplied in file format by the device manufacturer. The MIBs for all of the devices to be monitored are installed into the NMS, allowing the remote monitoring and management of the device. There are presently three versions of the SNMP protocol: SNMPv1, SNMPv2, and the latest and most secure SNMPv3. It is strongly advised to turn off SNMP protocol all together if it is not going to be used to manage a UPS. The default setting for most UPS manufacturers SNMP/HTTP agents has SNMPv1 selected and active. SNMPv1 supports a minimal single level password. Typically, agents are shipped from the UPS manufacturers with a default password making unauthorized access through SNMP child's play. SNMPv3 security supports hashing algorithms for secure multi-level password protection in addition to full data encryption, supported using differing shared keys. SNMPv3 should also be configured to limit access to one or two management workstation IP addresses and exclude all other addresses. This is usually the same IP addresses as the assigned SNMP trap receivers.

Hypertext Transfer Protocol (HTTP) with regards to a UPS SNMP/HTTP agent is a very unsecure protocol incorporating a couple of unencrypted, single-level user logins and passwords. It allows access to the main menu of the agent by anyone entering the agent's IP address into a web browser's URL line. It is recommended to turn off the HTTP port if the protocol will not be used. Hypertext Transfer Protocol Secure (HTTPs) is the preferred choice when web browser access is desired. HTTPs incorporate HTTP with SSL/TLS security. Even with the added security, it is suggested to use HTTPs in conjunction with a Remote Authentication Dial In User Service (RADIUS) server and enable the RADIUS support on the agent. RADIUS can effectively limit access to Internet, wired, and wireless networks.

Universal Plug and Play (UPnP) protocol is primarily used to support automatic configuration of residential networks devices. It should always be turned off in the UPS agents, as it allows easy detection of the agents configured on a network. It poses a real security risk in corporate or governmental networks.

Simple Mail Transfer Protocol (SMTP) supports the e-mail transmission over Internet Protocol and is used to send e-mail messages from the UPS agent upon specified UPS detected events. When configured, it can send e-mail to IT staff after normal working hours, in the event of a critical situation like a UPS failure. Again, it is best enabled in conjunction with RADIUS to provide additional security.

It is convenient for IT staff to use the Ping command to determine if a device is communicating with the network. Unfortunately, pinging for computers and devices over the Internet is the hacker's first act in finding unsecured computer and network devices. It is then a simple matter of entering the detected IP address into a "whois" domain look up on the Internet to determine what corporation or government agency the IP address is registered. They next run port scans of the IP address to determine if there are any open ports and their port number. The port number will indicate the type of communications port they have accessed. They next communicate with the port in its own language and attempt to gain access. This entire process is over in minutes. Once a device has been properly configured on a network, the Ping Echo support must be turned off in the device. Without a ping response from the agent, in addition to unused ports being turned off, it assures the agent is more stealthy and harder to detect on a network.

User Diagram Protocol (UDP) is used primarily by the UPS SNMP/HTTP agent to facilitate remote real-time firmware updates. It is a very unsecure protocol and should be turned off when not needed unless RADIUS is configured where UDP must be turned on.

In conclusion, it is essential to understand the security features and versions available in a specific UPS SNMP/HTTP agent as they can differ widely depending on the UPS manufacturer. Some manufacturers may not support IPv6, while others may not support SSHv2, SNMPv3, or RADIUS. UPS agent security features alone may not yield the level of security desired without the ability to turn off unused communications ports. The overall security of data and SCADA networks requires the careful selection of network-enabled devices, meticulous IT procedures, along with vigilant IT and network security departments. The UPS is a vital part of a resilient SCADA network; however, the SNMP/HTTP agent option selected may be critical to the continued reliability and safety of your process control. The level of network security required must be weighed against the level of security demanded by the application.

The UPS is a vital part of a resilient SCADA network; however, the SNMP/HTTP agent option selected may be critical to the continued reliability and safety of your process control.

ABOUT THE AUTHOR

Michael A. Stout, vice president of Engineering, Irwindale, Calif.-based Falcon Electric, Inc. (www.falconups.com), is an authority in commercial and military computer, power conversion and UPS systems, having more than two decades of experience. Stout specifies new UPS and power system implementation and design.

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Editor's note

As common as flow measurements are using orifice plates, there are various thoughts regarding design, application, rules of thumb, and field practice. Factors that can be considered include measurement errors as % full scale, % rate, bias error, ambient temperature induced errors (largely corrected by smart transmitters), and signal-to-noise ratio deteriorating at low flow rates. *InTech* invites other thoughtful insights on the subject.

Sizing orifice plates

Meeting modern expectations

By Allan G. Kern, P.E.

Orifice plates with differential pressure (DP) transmitters remain the workhorses of fluid flow measurement in the process industries, due to their proven robustness, ease of use, adaptability to a broad spectrum of applications, familiarity, and economy. The weak side of orifice plates, where otherwise properly applied and installed, is limited turndown, with a nonlinear loss of accuracy at lower flow rates due to the square-root nature of the flow/DP relationship.

With modern instrumentation and today's more stringent demands regarding material balances, yield and loss accounting, energy management, environmental reporting, and safety systems, users have developed greater expectations and requirements regarding accuracy of their flow measurement systems.

When sizing orifice plates, some new rules of thumb can be applied to significantly improve orifice plate turndown and accuracy, while gaining extended measurement range, in most applications. This can be accomplished for the cost and effort of revising the calculation, buying a new orifice plate, and re-configuring the transmitter, activities that are routinely carried out in any case.

FAST FORWARD

- Orifice plates remain the workhorses of fluid flow measurement.
- Updated sizing guidelines can bring big increases in accuracy and turndown.
- Capture the capabilities of smart transmitters and Fieldbus in orifice sizing.

Sources of error

There are many potential sources of error in orifice plate flow measurement. Many of them have been minimized in today's world or are outside of our control, such as variations in pipe diameter, orifice plate machining tolerances, and standardized flange taps. Modern DP transmitters have high accuracy (ca. 0.1%). The greatest sources of error today will come from temperature deviation from design (for liquids) and temperature, pressure, or specific gravity

deviations (for gases). The best practice where these parameters vary from design values is online compensation, utilizing built-in control system functions.

The remaining most common source of error is DP measurement error, whether due to transmitter inaccuracy, static pressure effects at high pressures, or imperfect field installation. The effect of measurement error can be greatly reduced by employing appropriate rules of thumb when "sizing" the orifice plate, i.e., when calculating the orifice size, differential pressure, and maximum flow.

An orifice plate field installation for gas flow. The low point (circled) can be expected to trap liquid and cause a small measurement error (ca. 1-inch). Although the rules for orifice plate installation are well-known, such non-ideal field conditions are common throughout industry. It can be good practice to assume an imperfect installation and size the orifice plate to minimize its effect.

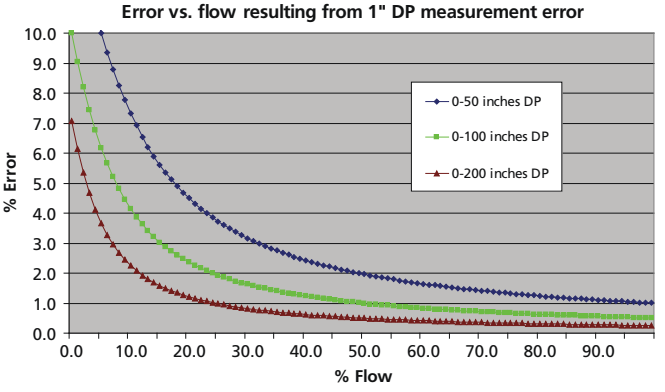
Selecting full-scale DP

The orifice flow measurement error figure shows the effect of a 1-inch DP measurement error on accuracy for three different full-scale DPs (50, 100, and 200 inches of water). The square root nature of the relationship amplifies the effect at lower flow rates, making it essential to avoid this operating region. One way to do this is to size the orifice for a greater full-scale DP, which moves the curves downward into the higher-accuracy region in the figure.

Based on an assumption of a potential 1-inch DP measurement error and a goal of less than 2% resultant error in flow (orifice plates are commonly considered "2% devices"), an orifice plate sized for 50-inches full-scale DP (a common design practice) meets this criteria only above 50% of flow, for a turndown of only 2:1. A full-scale DP of 100 inches (the most common design practice today) meets this criteria above 25% of flow, for a turndown of 4:1. And a full-scale DP of 200 inches (an uncommon practice today) meets this criteria above 10% of flow, for a turndown of 10:1.

Faced with today's more stringent performance expectations, what does the figure say about reducing this error from 2% to 1%? An orifice plate sized for 50-inches full-scale DP only meets this requirement at near full-scale flow (>90%). A full-scale DP of 100 inches only meets this above 50% of flow, or a 2:1 turndown. And a full-scale DP of 200 inches meets these criteria down to 25% of flow, for a turndown of 4:1.

For any given flow, any of these choices is most likely completely acceptable and would likely go unscrutinized, i.e., in most cases, an orifice can be sized for anywhere from 50 to 200 inches full-scale DP, while staying well within the beta ratio and other guidelines.



Orifice flow measurement error resulting from a 1-inch differential pressure (DP) measurement error for various full-scale DPs. Error is expressed as percentage of full-scale flow.

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Consequently, based on an often arbitrary choice, turndown can vary from 2:1 to 10:1, and accuracy from 4% or more to 1% or less.

Selecting maximum flow rate

Another often somewhat arbitrary choice in orifice sizing is the maximum flow rate. As this discussion shows, selecting an unnecessarily high maximum flow rate will compromise accuracy at lower (normal) flow rates, so selecting a maximum flow rate based on infrequent conditions carries an accuracy penalty under normal conditions and should be avoided to the extent possible.

Many users do not realize that with modern smart transmitters, which are *configured* by the end user, not *calibrated,* the maximum flow measurement limitation is the upper range limit (URL) of the transmitter (often 200 to 500 inches, depending on the make and model), not the configured upper range value (URV), which is the orifice sizing full-scale DP. This removes the incentive to increase the full-scale DP in order to capture infrequent high flow conditions, since the limitation is the transmitter URL, not the configured URV.

Taking advantage of this can have subtle and initially confusing implications on traditional 4-20 milliamp analog input systems, but on digital systems, such as Fieldbus, the practice is simply to configure the control system "high" scale, or URV, equal to the orifice sizing full-scale DP, and set the control system "extended" scale based on the URL of the transmitter. This practice allows the orifice sizing full-scale DP to be chosen appropriately for normal conditions, thereby maximizing accuracy, while taking advantage of the full measurement range of the transmitter to capture infrequent high flow rates.

Modern safety systems also create increased incentive for orifice plate accuracy. Safety system transmitters are traditionally given a reduced range in order to improve accuracy around the trip setting. But in modern safety systems, design calls for the safety transmitters to have the same range as the control system transmitter in order to provide diagnostic discrepancy alarms.

Caveats

There are a few caveats to shrinking the orifice and increasing the DP in order to improve accuracy, but they are not usually significant. As mentioned, the beta ratio, which is the ratio of orifice diameter to inside pipe diameter, should remain within the established design range of ca. 0.3 to 0.7. The table shows increasing the full-scale DP from 50 to 200 inches will decrease the beta ratio from ca. 0.60 to 0.45, still well within range on both ends.

Full-scale DP	Beta ratio	Increase in permanent pressure drop
50 inches	0.60	
100 inches	0.52	1.6 PSIG
200 inches	0.45	3.2 PSIG

Effect of full-scale DP on beta ratio and permanent pressure drop

Second, there can be an energy penalty for increased permanent pressure loss, which is typically 50–90% of DP. This amounts to ca. 3 PSIG additional loss when switching from 100 to 200 inches full-scale DP, when at full-scale flow. In most cases, this is not significant, and the pressure is lost elsewhere in the process, for example, across a control valve.

Third, with a higher DP, there is the possibility of cavitation or flashing in liquid service. This is not usually an issue and is typically flagged by the orifice sizing software.

New rules of thumb

For greater orifice plate accuracy and turndown, use a larger full-scale DP. Consider using 200 inches as a default, rather than 50 or 100 inches.

Avoid selecting an unnecessarily high maximum flow for sizing. Utilize the capabilities of modern smart transmitters to capture infrequent high flow rates.

Use the figure to gauge if expected accuracy and turndown are satisfactory, or if improvements could be easily captured by selecting a higher full-scale DP.

ABOUT THE AUTHOR

Allan Kern (allan.kern@yahoo.com) has 30+ years of experience in process control and has authored numerous papers on multivariable control, inferentials, decision support systems, safety instrumented systems, distillation control, and other topics, with an emphasis on practical process control effectiveness. Kern is a professional control systems engineer in California, a senior member of ISA, and a member of the *InTech* editorial advisory board.

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A China perspective for 2012

By John Erskine III

When the subject of China comes up, I often ask people if they remember that 1970's Calgon detergent commercial: Mr. Lee's customer asks him how he gets the laundry so clean; his sly reply: "Ancient Chinese secret!" The ad depicts stereotypes that simply would not get air time today. Yet, when reviewing recent publications or forums on doing business in China, one finds plenty of comments that sound less like actionable advice, and more like ancient Chinese secrets:

- Discussion threads on Asia/Pacific business, where posts make mention of China's invention of the umbrella and gunpowder
- Experts on China business who emphasize the country's 5,000-year history
- And a favorite: The admonishment that Westerners must learn to discern "when yes really means no." Do Chinese authors warn their readers to "beware the Socratic Method?"

It is not that these observations are wrong, but I remain surprised at the relevance that gets ascribed to them in the context of doing business with Chinese companies today. For the executive whose managers travel to China for growing sales or for establishing sources of supply, a more practical approach is warranted. An approach focused on identifying similarities and aligned goals, tends to work in China as it does at home.

For a small or mid-size manufacturer, doing business in China today has never been easier. When we meet with dealers, it is more common to see a corporate mission statement in the office. Meetings are focused on sales development the way Western-trained managers understand it: where the dealer sees his role in the value chain as providing market data, appointing product champions, expanding geographic coverage, and investing in local service and support capabilities. Today, it is common to meet Chinese managers with business degrees from the U.S. or Canada. With a baseline relationship established with a solid dealer who truly *speaks our language,* it means we can focus energies on strategic goals—growing market share and offering a product suite with the broadest appeal to the local Chinese market.

Another observation: Today when visiting in China, we are not lectured on the "importance of *guanxi.*" (*Guanxi xue* involves the exchange of gifts, favors, and banquets; the cultivation of personal relationships and networks of mutual dependence.) It is not that this idea has disappeared from Chinese culture, but perhaps more managers there have found other ways to grow their business. It is telling that a company like Taiwan Semiconductor (TSMC) has written in its corporate core values statement, a rejection of "connections or *guan-xi*" in its hiring and business practices. So things are changing—that's not news, but for those still contemplating working with Chinese companies, what used to seem like challenges specific to China, today seem much more like typical business challenges we face at the home office.

On the supply side, for companies who are not considering China-based operations, today, one finds many Chinese companies who will act as an intermediary between Western and Chinese manufacturers. These brokers will take a set of drawings and specifications, and find local vendors with the technical capabilities, quality control, and capacity, to fit their Western partner's requirements. They handle all communications and negotiations with vendors, monitor vendor on-time delivery and quality, and often have their own engineering staff and Q.C. labs. Companies like this come to the table with an answer to the negative perceptions of Chinese quality, and with a solution for companies lacking the resources to go it alone. The reality today is we get excellent quality from our Chinese sources—though as at home, the process requires monitoring and management.

For anyone in charge of international sales or sourcing, remember your managers still have a job to do: setting expectations with suppliers; finding common ground with sales partners; working a game plan. Consider, when it comes to China, are your managers at risk of perpetuating their own stereotypes? Is potential China business getting shortchanged at your company because of what managers think they know about it? In China, as in other overseas markets, insist managers remain focused on the business and business goals, and that they work at clear communications with their foreign counterparts. Know small and mid-size companies have more options than ever, to help ensure an investment in China pays off. In the long run, managers will tend to find that while some "secrets" may persist, they need not be a barrier to success.

ABOUT THE AUTHOR

John Erskine (johnerskineiii@racinefed.com) is a vice president at Racine Federated Inc., manufacturer of flowmeters of various technologies including ultrasonic, turbine, vortex-shedding, differential pressure, and variable area.

PID tuning rules

By Greg McMillan

Safety, equipment and environmental protection, process efficiency and capacity, product quality, and control system maintenance depend on PID tuning.

Fast Forward

- Peak error, integrated error, and rise time are inversely proportional to PID gain.
- A unified equation for PID gain is applicable to all major types of processes.
- PID features can inherently prevent cycling without retuning.

Nearly every automation system supplier, consultant, control theory professor, and user has a favorite set of PID tuning rules. Many of these experts are convinced their set is the best. A handbook devoted to tuning has over 500 pages of rules. The enthusiasm and sheer number of rules is a testament to the importance of tuning and the wide variety of application dynamics, requirements, and complications. The good news is these methods converge for a common objective. The addition of PID features, such as setpoint lead-lag, dynamic reset and output velocity limits, and intelligent suspension of integral action enable the use of disturbance rejection tuning to achieve other system requirements, such as maximizing setpoint response, coordinating loops, extending valve packing life, and minimizing upsets to operations and other control loops.

Potential performance

The purpose of a control loop is to reject undesired changes, ignore extraneous changes, and achieve desired changes, such as new setpoints. PID control provides the best possible rejection of unmeasured disturbances (regulatory control) when properly tuned. The addition of a simple deadtime block in the external reset path can enhance the PID regulatory control capability more than other controllers with intelligence built-in to process dynamics, such as model predictive control. In plants, unknown and extraneous changes are a reality, and the PID is the best tool if properly tuned. The test time has been significantly reduced for the most difficult loops. Simple equations have been developed to estimate tuning and resulting performance for a unified approach. (Equation derivations and a simple tuning method are in the online version.)

Control requirements

The foremost requirement of a PID is to prevent the activation of a safety instrumentation system or a relief device and the prevention of an environmental violation (RCRA pH), compressor surge, and shutdown from a process excursion. The peak error (maximum deviation from setpoint) is the most applicable metric. The most disruptive upset is an unmeasured step disturbance that would cause an open loop error (E_o) if the PID was in manual or did not exist. The fraction of open loop error seen in feedback control is more dependent upon the controller gain than the integral time since the proportional mode provides the initial reaction important for minimizing the peak error. Equation (1) shows if the product of the controller gain (K_c) and open loop gain (K_o) is much greater than one, the peak error (E_x) is significantly less than the open loop error. The open loop gain (K_o) is the product of the final element, process, and measurement gain and is the percent change in process variable divided by the percent change in controller output for a setpoint change. For most vessel and column temperature and pressure control loops, the process rate of change is much slower than the deadtime. Consequently, the controller gain can be set large enough where the denominator becomes simply the inverse of the product of the gains. Conversely, for loops dominated by deadtime, the denominator approaches one, and the peak error is essentially the open loop error.

$$(1) \quad E_x = \frac{1}{(1 + K_c * K_o)} * E_o$$

The peak error is critical for product quality in the final processing of melts, solids, or paste, such as extruders, sheet lines, and spin lines. Peak errors show up as rejected product due to color, consistency, optical clarity, thickness, size, shape, and in the case of food, palatability. Unfortunately, these systems are dominated by transportation delays. The peak errors and disruptions from upstream processes must be minimized.

The most widely cited metric is an integrated absolute error (IAE), which is the area between process variable and the setpoint. For a non-oscillatory response, the IAE and the integrated error (IE) are the same. Since proportional and integral action are important for minimizing this error, Equation (2) shows the IE increases as the integral time (T_i) increases and the controller gain decreases.

$$(2) \quad E_i = \left[\frac{(T_i + \Delta t_x + \tau_f)}{K_o * K_c} \right] * E_o$$

Equation (2) also shows how the IE increases with controller execution time (Δt_x) and signal filter time (τ_f). The equivalent deadtime from these terms also decreases the minimum allowable integral time and maximum allowable controller gain, further degrading the maximum possible performance. In many cases, the original controller tuning is slower than allowed and remains unchanged, so the only deterioration observed is from these terms in the numerator of Equation (2). Studies on the effect of automation system dynamics and innovations can lead to conflicting results because of the lack of recognition of the effect of tuning on

the starting case and comparative case performance. In other words, you can readily prove anything you want by how you tune the controller.

IE is indicative of the quantity of product that is off-spec that can lead to a reduced yield and higher cost ratio of raw material or recycle processing to product. If the off-spec cannot be recycled or the feed rate cannot be increased, there is a loss in production rate. If the off-spec is not recoverable, there is a waste treatment cost.

A controller tuned for maximum performance will have a closed loop response to an unmeasured disturbance that resembles two right triangles placed back to back. The base of each triangle is the total loop deadtime and the altitude is the peak error. If the integral time (reset time) is too slow, there is slower return to setpoint. If the controller gain is too small, the peak error is increased, and the right triangle is larger for the return to setpoint.

Process dynamics

The major types of process dynamics are differentiated by the final path of the open loop response to a change in manual controller output assuming no disturbances. (The online version shows the three major types of responses and the associated dynamic terms.) If the response lines out to a new steady state, the process is self-regulating with an open loop time constant (τ_o) that is the largest time constant in the loop. Flow and continuous operation temperature and concentration are self-regulating processes. If the response continues to ramp, the process is integrating. Level, column and vessel pressure, batch operation temperature, and concentration are integrating processes. If the response accelerates ,reaching a point of no return, the process has positive feedback leading to a runaway. Batch or continuous temperature in highly exothermic reactors (e.g., polymerization) can become runaway processes. Prolonged open loop tests are not permitted, and setpoint changes are limited. Consequently, the acceleration is rarely intentionally observed.

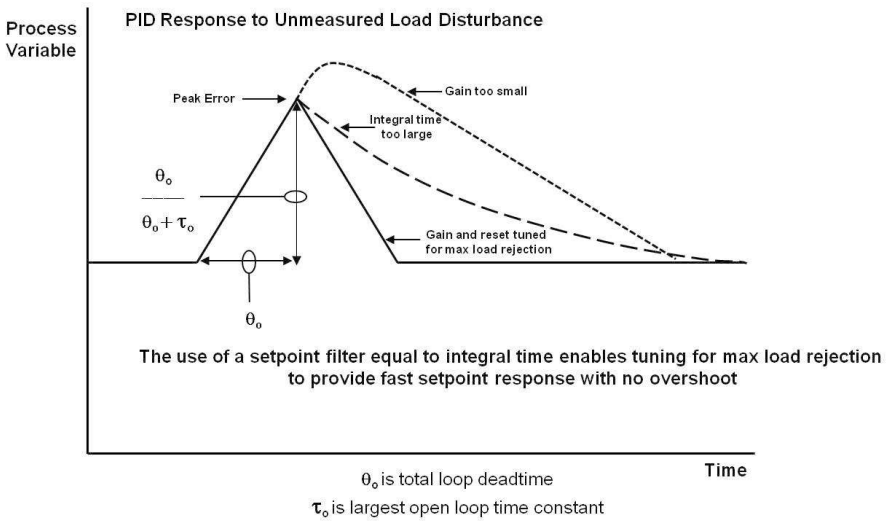
Unified approach

The three major types of responses have an initial period of no response that is the total loop deadtime (θ_o) followed by the ramp before the deceleration (inflection point) of a self-regulating response and the acceleration of the runaway response. The percent ramp rate divided by the change in percent controller output is the integrating process gain (K_i) with units of %/sec/%, which reduces to 1/sec.

For at least 10 years, slow self-regulating processes with a long time to deceleration have shown to be effectively identified and tuned as "near integrating" or "pseudo integrating" processes, leading to a "short cut tuning method" where only the deadtime and initial ramp rate need to be recognized. The tuning test time for these "near integrating" processes can be reduced by over 90% by not waiting for a steady state. Recently, the method was extended to runaway processes and to deadtime dominant self-regulating processes by the use of a deadtime block to compute the ramp rate over a deadtime interval. Furthermore, other tuning rules were found to give the same equation for controller gain when the performance objective was maximum unmeasured disturbance rejection. For example, the use of a closed loop time constant (λ) equal to the total loop deadtime in Lambda tuning yields the same result as the Ziegler Nichols (ZN) ultimate oscillation and reaction curve methods if the ZN gain is cut in half for smoothness and robustness. Equation (3) shows the controller gain is half the inverse of the product of integrating process gain and deadtime.

$$K_c = \frac{0.5}{K_i * \theta_o} \quad (3)$$

The profession realizes that too large of a controller gain will cause relatively rapid oscillations and can instigate instability (growing oscillations). Unrealized for integrating process is that too small of a controller gain can cause extremely slow oscillations that take longer to decay as the gain is decreased. Also unrealized for a runaway process is that a controller gain set less than the inverse of the open loop gain causes an increase in temperature to accelerate to a point of no return. There is a window of allowable controller gains.



The effect of integral time on the maximum possible disturbance rejection

Also realized is too small of an integral time will cause overshoot and can lead to a reset cycle. Almost completely unrealized is that too slow of an integral time will result in a sustained overshoot of a setpoint that gets larger and more persistent as the integral time is increased for integrating processes. Hence a window of allowable integral times exists. Equation 4a provides the right size of integral time for integrating processes. If we substitute Equation 3 into Equation 4a, we end up with Equation 4b, which is a common expression for the integral time for maximum disturbance rejection. Equation 4a is extremely important because most integrating processes have a controller gain five to 10 times smaller than allowed. The coefficient in Equation 4b can be decreased for self-regulating processes as the deadtime becomes larger than the open loop time constant (τ_o) estimated by Equation 5.

(4a) $$T_i = \frac{2}{(K_c * K_i)}$$

(4b) $$T_i = 4 * \theta_o$$

(5) $$\tau_o = \frac{K_o}{K_i}$$

The tuning used for maximum load rejection can be used for an effective and smooth setpoint response if the setpoint change is passed through a lead-lag. The lag time is set equal to the integral time, and the lead time is set approximately equal to ¼ the lag time.

For startup, grade transitions, and optimization of continuous processes and batch operations, setpoint response is important. Minimizing the time to reach a new setpoint (rise time) can in many cases maximize process efficiency and capacity. The rise time (T_r) for no output saturation, no setpoint feedforward, and no special logic is the inverse of the product of the integrating process gain and the controller gain plus the total loop deadtime. Equation 6 is independent of the setpoint change.

(6) $$T_r = \frac{1}{(K_i * K_c)} + \theta_o$$

Complications, easy solutions

Fast changes in controller output can cause oscillations from a slow secondary loop or a slow final control element. The problem is insidious in that oscillations may only develop for large disturbances or large setpoint changes. The enabling of the dynamic reset limit option and the timely external reset feedback of the secondary loop or final control element process variable will prevent the primary PID controller output from changing faster than the secondary or final control element can respond, preventing oscillations.

Aggressive controller tuning can also upset operations, disturb other loops, and cause continual crossing of the split range point. Velocity limits can be added to the analog output block, the dynamic reset limit option enabled, and the block process variable used as the external reset to provide directional move suppression to smooth out the response as necessary without retuning.

The different closed loop response of loops can reduce the coordination, especially important for blending and simplification of the identification of models for advanced process control systems that manipulate these loops. Process nonlinearities may cause the response in one direction to be faster. Directional output velocity limits and the dynamic reset limit option can be used to equalize closed loop time constants without retuning.

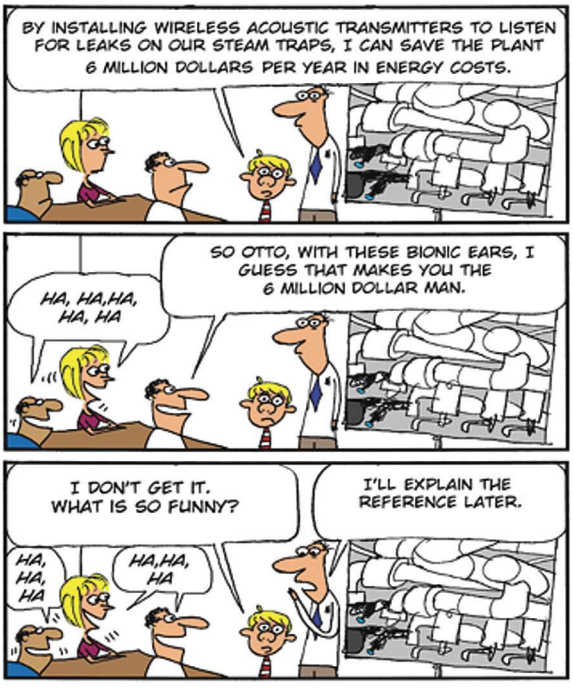
Final control element resolution limits (stick-slip) and deadband (backlash) can cause a limit cycle if one or two or more integrators, respectively, exist in the loop. The integrator can be in the process or in the secondary or primary PID controller via the integral mode. Increasing the integral time will make the cycle period slower but cannot eliminate the oscillation. However, a total suspension of integral action when there is no significant change in the process variable and when the process is close to the setpoint can stop the limit cycle. The output velocity limits can also be used to prevent oscillations in the controller output from measurement noise exceeding the deadband or resolution limit of a control valve preventing dither, which further reduces valve wear.

Bottom line

Controllers can be tuned for maximum disturbance rejection by a unified method for the major types of processes. PID options in today's DCS, such as setpoint lead-lag, directional output velocity limits, dynamic reset limit, and intelligent suspension of integral action, can eliminate oscillations without retuning. Less oscillations reduces process variability, enables better recognition of trends, offers easier identification of dynamics, and provides an increase in valve packing life.

ABOUT THE AUTHOR

Greg McMillan (Greg.McMillan@Emerson.com) is a retired Senior Fellow from Solutia-Monsanto and received the ISA Life Achievement Award in 2010. Presently, he is a principle consultant with CDI Process & Industrial contracting at Emerson Process Management in DeltaV R&D. He is also a part-time employee of MYNAH Simulation Technologies in Saint Louis.



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Getting involved with an ISA section

By Mat Merten

In January 2009, I became a licensed engineer. After the excitement wore off, I became concerned about how to get all my needed professional developmental hours (PDH). Although a member of ISA, I was not involved in the local section in Oak Ridge, Tenn., nor had I heard about anything going on with the section. I reached out to some of the local officers I found on the website and said I wanted to get involved. After meeting with the officers, we decided to kick-start the Oak Ridge Section and have monthly technical meetings. I volunteered to be the program chair, which essentially meant I became the face of the section.

The first few meetings were rough. Few people came, and the section was going through a lot of cash. However, we kept at it, and after a year, we got momentum going in the community, and now have upwards of 40 participants. Speakers have started reaching out to us because they know they will have a captive audience. We are really proud of how far the Oak Ridge Section has come.

I have gained a tremendous amount, professionally and more important personally, from being involved in technical societies.

Since this section began, I have personally gotten involved in FIRST Robotics, thanks to having attended ISA Automation Week 2010, where the keynote speaker, FIRST founder Dean Kamen, sparked my interest. A few months after that event, I was notified by the National Society of Professional Engineers (NSPE), another technical society that I am a member of, that a local regional competition of FIRST Robotics was going to be held in Knoxville, Tenn. In just a few days, I convinced the regional director that since I helped run the local section of ISA, I would be an excellent choice for a judge. The director agreed.

A judge's commitment is Thursday evening and all day Friday and Saturday. After spending 30 seconds on the game floor Thursday evening, I knew this event was an incredible opportunity and immediately thought it would be perfect for Amber Hodge, a marvelous teacher I previously had met while judging a science fair at a local Islamic school, Annoor Academy. (I learned about the science fair through yet another organization—the American Nuclear Society.) I was not able to contact Hodge until late Friday night, but sent her an e-mail anyway telling her she had to come and see it herself. My expectations were low that she would come since it was so last minute, but she said she would be there. We met the next day, and I showed her around the floor and pits. At that moment, she was hooked, and we both committed to starting a team at Annoor.

Starting a FIRST team is tough, but after many long meetings, parental discussions, fundraising (yes, the ISA Oak Ridge Section is an official sponsor), table building, field trips, and robot troubleshooting, we have a team we literally could not be more proud of. Only in elementary school, the children have learned and come together as a team beyond our wildest expectations. We have appointed a Team Captain, Hardware Captain, Software Captain, and Project Captain. These children have already learned about many valuable technical topics from motors, power, trigonometry, to even food safety, which was the theme the kids had to research in 2011. Outside the world of the team, the kids have more self-confidence with their new technical prowess, and they are envied by their classmates who are not on the team. Each has taken incredible ownership, and we are so proud of them.

Two main points: I have gained a tremendous amount, professionally and more important personally, from being involved in technical societies. Being a coach of a robotics team has been simply a wonderful experience, and there is no way this chain of events would have occurred if not for ISA and working with many other people. Second, the local ISA Sections can be a force of good in the community, not only from a technical and business knowledge sharing standpoint, but also from a giving back to the community standpoint.

The Oak Ridge Section may not be the best ISA section out there, but it is improving. Our goals for 2012 are to have a high-quality website, host a vendor's fair, present 10 technical presentations, and my personal favorite, become the official sponsors of multiple FIRST Robotics teams throughout the community. We are getting there and encourage others to get involved with their local ISA sections as well.

ABOUT THE AUTHOR

Mat Merten, P.E., is the Federal/Nuclear Engineering manager for BESCO in Oak Ridge, Tenn. He is involved in various organizations throughout the community and is the most passionate about his Robot Kids.

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Path set for 2012

By Robert E. Lindeman

2011 was a busy year for me as the ISA President-elect and went by quickly. I was honored to attend multiple ISA events, including division symposia, summits, leadership conferences, section meetings, and numerous web meetings and teleconferences. These events opened opportunities for me to interact with hundreds of ISA members and witness the amazing depth and breadth of knowledge they have to offer the automation community. As a Society, we are indeed setting the standard for automation throughout the world.

As your 2012 ISA President, my main focus will be to provide tangible benefits for ISA members and to be instrumental in the continued advancement of the practice of automation. My focus aligns with the Society's 2012–2015 strategic goals:

- Serve students, professionals, and industry by delivering knowledge resources via publications, conferences and exhibits, training, and other programs.
- Develop globally recognized standards, publications, and certifications for the automation community.
- Attract and retain automation professionals as members and customers worldwide.

To make these goals a reality, we must continue to provide current technical content to support ISA's technical conferences and events, actively support divisions, strengthen sections, and offer exceptional programs to automation professionals.

ISA is committed to enhancing its position as the source for authoritative standards and practices in all automation areas. We can benefit from initiating discussions with other professional organizations to understand how to make access to ISA standards for their members more applicable to them. For example:

- Collaborating with the leaders of automation companies and educational providers to encourage wider acceptance and usage of ISA standards as part of their culture
- Working with political entities to raise awareness of the availability and applicability of ISA standards to government procurement policies

ISA offers excellent certification programs, but they do not have wide-spread acceptance throughout the automation profession. I plan to establish a special task force to market these programs to industry, government, and educational institutions. I would like to reach the point where holding an ISA certification becomes a standard requirement in the automation field, much like the Project Management Professional (PMP) has become for Project Management.

In the coming years, the Society has a number of training and educational goals. We will continue expanding our effort to establish community college, undergraduate, and graduate degree programs in Automation Studies as part of the Engineering Curriculum. ISA will push forward with providing flexible, affordable training programs that best serve individual and company needs. The Society will also look at how companies are training their automation engineers and technicians and discuss ways to implement its own programs that will augment companies' existing training.

Last, but certainly not least, ISA Web 2.0 needs to be a portal to the technical world, while maintaining trusted content. Last summer, the first steps were taken with the roll out of *ISA Interchange,* and a larger conversation began between automation professionals on vehicles such as Twitter, Facebook, and LinkedIn. ISA anticipates the technical posts and conversations will continue to thrive going forward.

Meeting our goals will take all of us working together. I urge you to contribute as an individual, section, district, division, department member, and leader to help strengthen ISA. As you develop plans and consider options in your role as an ISA volunteer, please keep these two questions in mind:

- How are we benefitting our members?
- How are we advancing the practice of automation?

I look forward to working with each of you in 2012 and encourage your input. Working together, we can set the standard for automation.

ABOUT THE AUTHOR

Robert E. Lindeman, CAP, PMP, is the 2012 ISA President.

Memoriam: Hugh Wilson

1978 ISA President Hugh Wilson passed away late last year. Wilson was 85. A memorial service was held in early January. Wilson was an icon of Moore Products Co. He started in Chicago as a young sales engineer and quickly moved to branch manager. He raised his hand and said he would take over DuPont, which was a stagnant account for Moore for many years. Hugh eliminated all sales credit for himself and started visiting DuPont sites teaching control theory courses at the sites. He started U.K. operations, then Canada, South Africa, Asia Pacific, and other areas of the world. Wilson retired as vice president, International Sales/Operations. Wilson held many important roles in ISA, none more important as ISA President. In all the years he was active in ISA, he never lost sight of the fact that he worked for Moore Products, and so many of his successes in setting up international operations was a result of his networking at ISA.

Memoriam: Ernie Magison

Ernest Carroll Magison was born 15 October 1926 and died on 3 December 2011 in Tampa, Fla. He was 85. He was a published author for ISA, authoring *Electrical Instruments in Hazardous Locations*. Magison was active in standards development at ISA, IEC, and NFPA for four decades. He has authored 40 articles, as well as papers and several books. Magison worked as an electrical engineer for Honeywell for 33 years and as a professor at Drexel University in Philadelphia for 15 years concurrently. He is survived by his wife of 28 years, Dixie, six daughters, and two stepdaughters.

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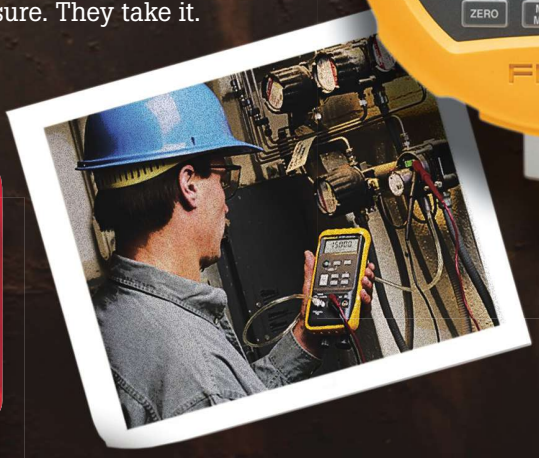

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Compliance with new pipeline rule offers opportunity to implement best practices

By Russel Treat

Many pipeline operators will spend substantial time and effort reworking their control rooms and systems to comply with the new Pipeline Hazardous Materials Safety Administration's (PHMSA) Control Room Management (CRM) rule by 1 August 2012. With a small amount of additional effort, pipeline operators can leverage compliance efforts into best practices that will serve them now and well into the future.

Why the new rule?

In investigating incidents, PHMSA found pipeline controllers (the person using the Supervisory Control and Data Acquisition [SCADA] system) may be qualified but not always successful in managing abnormal situations or events. The controller's ability to manage abnormal situations can be influenced by ineffective procedures, fatigue, or even limitations in the SCADA system itself.

To comply with the CRM rule, pipeline operators must:

- Provide effective operating and maintenance procedures, including specific requirements for training
- Match the control room environment and equipment (including displays, alarming, environment, etc.) to human capability
- Provide controllers with warnings and guidance when abnormal operations occur

The requirement to "match to human capability" presents one of the greatest challenges to pipeline operators, demanding solutions and systems that differ significantly from anything they have in their control rooms today.

Situational awareness, permission to operate

Matching a control room to human capability is the science of human factors. The idea is to provide the information required to maintain pipeline safety and operational control without too much workload or confusion. Current control rooms are generally designed for normal operations. However, it is during abnormal or emergency operations when proper human factors engineering can make the difference between success and failure.

Fundamental to success are two key concepts: "situational awareness" and "permission to operate." Originally a battlefield science, *situational awareness*, as applied to control systems, refers to a single display that quickly provides the operating condition of the pipeline: normal, abnormal, or emergency.

The second key concept in our approach is permission to operate. The premise is you maintain permission to operate as long as you maintain situational awareness (i.e., you understand the operating condition). When you lose situational awareness, then you no longer have permission to operate and need to return to a safe operating condition. This could require shutting down or changing operating parameters, depending on the process.

Frequently, a process moves from normal operations into abnormal operations before the controller receives any notification. The idea is to design displays to show abnormal operations before an alarm is required. Then, use alarms levels to ensure the operator has time to properly identify the cause, determine the solution, take corrective action, and allow for the process to respond.

Our recommended best practice is to define *alarm* as something requiring controller action. Further, EnerSys recommends pre-engineering all alarms and making the results available to improve the controller's ability to accurately respond. Because field personnel may be required to drive to remote sites to handle manual portions of the process, planning for elapsed time is critical to success.

Retaining permission to operate requires implementing alarm management in such a way that alarms are clear and unambiguous. Systems must be designed so the controller will not act too early when a problem arises but has sufficient time to diagnose and determine what specific action is required, with enough of a cushion for the process to respond.

Reconsidering SCADA

Historically, we think of SCADA as communications and wiring bringing signals to a display. As an industry, we have not given much thought to the operator interface. SCADA is a complex toolset, including multiple software components and communications technologies, complex networking, and field equipment. To have a firm grasp on the entire system, the pipeline operator needs a well-documented data flow and infrastructure.

What often is missing in these systems is a way to record shift handover and controllers' logs, as well as processes for alarm management, simulation training, leak detection modeling, commissioning support, and change management. In addition, some rework—in the way data is displayed on the HMIs and the way the alarms are processed—is needed. Some of these changes may be strictly procedural, some may entail software updates, and some may require a combination of both.

Compliance is opportunity

Addressing this new rule offers the pipeline operator a prime opportunity to take a structured approach to best practices that will significantly improve their operations. The requirements will drive a rework of displays, alarms, and procedures. Doing this incrementally will require a very thoughtful approach to the disciplined methodology.

ABOUT THE AUTHOR

Russel Treat (rtreat@enersyscorp.com) is founder and president of EnerSys Corporation, a Control System Integrators Association Certified member.

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Change needed in pinch-valve standard definitions

By Gerald Liu

Today's only existing pinch-valve standards (from ISA) capture years of success to save time and cost for instrument engineers and designers who use these standards in projects related to pinch-valve design, engineering, construction, installation, commissioning, and maintenance. Pinch-valve standards can increase efficiency, productivity, and reliability in plant operations and provide safety to the users and their equipment in application of pinch valves. Yet there is some discrepancy between the existing definitions in ANSI/ISA-75.05.01, *Control Valve Terminology,* and the draft standard ISA-75.10.03, *Installed Face-to-face Dimensions for Shell and Tube Flanged Pinch Valves.* Here is a history and an explanation of the need for change.

Pinch-valve uses

Historically, the rubber/elastomer sleeve (bladder) and lining allowed pinch-valve manufacturers more freedom with elasticity and isolation to design pinch valves and create face-to-face dimensions for their valves than those manufacturers of conventional control valves.

Elasticity is a property of the rubber/elastomer that allows bubble-tight sealing and shut-off, as well as some flexibility in designing the overall valve body length. The term "soft face-to-face dimension" refers to the linear dimension from the far left side of one flange of a pinch valve to the far right side of the other flange before installation of the valve into the piping system, i.e., when the rubber/elastomer sleeve and the elastic lining on the flanges have not been compressed.

Isolation is not a property of the rubber/elastomer, but manufacturers can achieve complete isolation of the process medium from the valve body with the rubber/elastomer sleeve in pinch valves. Certain pinch-valve designs have mesh wires (sensors) built inside the sleeve for leak-detection in the process. When the sleeve starts to break, such sensors monitor early leaks before breaks are complete.

Another reason pinch valves have their own face-to-face dimensions is they handle fluids conventional control valves could not handle, such as corrosives (acids and caustics used in water treatment/chemical plants), slurries (from sewage systems), and abrasive solids (coarse sands in mining).

Standardization

There was no standard stipulating what rubber/elastomer material to use for specific fluids in the 1950s; only guidelines existed. While some of the properties overlap each other, pinch-valve manufacturers have their own successful, but untold, stories related to solving unique problems. Pinch valves came into being when engineers saw conventional control valves could not handle corrosives, slurries, and abrasive solids. The engineers worked on their own drawing boards, selected their own rubber/elastomers, and built their own valves. Yet not all pinch-valve manufacturing achieved success, and some pinch valves disappeared from the market. This is why there are five columns of face-to-face dimensions in the current draft standard ISA-75.10.02, *Installed Face-to-Face Dimensions for Dual Pinch Flanged Clamp or Pinch Valves (Classes 125 and 150),* and four columns of face-to-face dimensions in ISA-75.10.03.

A standard of face-to-face dimensions can have various sets of dimensions, such as ASME/ANSI B16.10, which has 20 different columns of face-to-face dimensions. Furthermore, having a standard does not mean the standard can never be changed.

Need for re-defining

The objective of having a standard is not the same as providing one fixed set of face-to-face dimensions for all pinch valves. A standard of face-to-face dimensions can have various sets of dimensions, such as ASME/ANSI B16.10, which has 20 different columns of face-to-face dimensions. Furthermore, having a standard does not mean the standard can never be changed.

One view is shell-and-tube pinch valves are on-off valves, without a physical actuator for their operation, and perhaps should not be considered as a control valve under the definition in ANSI/ISA-75.05.01.

Another view, that of the ISA75.10 subcommittee, is control valves and actuators need to be redefined in broader terms. First, on-off control is a type of process control. Second, current literature illustrates how shell-and-tube pinch valves can be used in throttling (modulating) control when equipped properly with transmitters and controllers, just like any control valve. Third, the definition of control valves in ANSI/ISA-75.05.01 is problematic. Under ANSI/ISA-75.05.01, a control valve is defined as having to have a physical actuator attached. This is not practical in industry. Pressure-reducing valves have no actuators, but they are still control valves. Similarly, shell-and-tube pinch valves are operated by a pressure signal; they work without an actuator and can perform modulating controls like a control valve. To broaden the application of control valves, the ISA75.05 subcommittee should revisit the definitions of the terms "control valves" and "actuators" in ANSI/ISA-75.05.01.

Thus, in ISA-75.10.03, control valves and actuators are re-defined as described above. Control valves are any valves operated by an instrument signal. Actuators are transducers for converting instrument signals to displacements. With these definitions in place, ISA75.10 can establish the new ISA-75.10.03.

ABOUT THE AUTHOR

Gerald Liu, P.E. (gerald.liu@shaw.ca) is the chairman of ISA75.10 and lives in Calgary, Alberta, Canada. If you are interested in joining the efforts of ISA75.10, contact him or Ellen Fussell Policastro (efussell@isa.org.)

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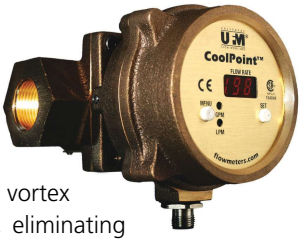
GF Piping Systems, www.gfpiping.com

Vortex shedding flowmeters



The CoolPoint vortex shedding flowmeters offer Intrinsic Safety as an option. CoolPoint vortex shedding flowmeters are electronic instruments that measure and monitor water flows. Unlike mechanical switches, the vortex shedding meters have no moving parts to stick or coat, eliminating potential for clogging by particles. The Intrinsic Safety feature is available on 1/4", 3/8", ½", 3/4", 1", 1 ½" and 2" CoolPoint meters. The transmitter with Intrinsic Safety is a 2 wire 4-20mA transmitter that has approved barriers and no display, alarm, or pulse output. It conforms to requirements for NEC Class I, Division 1, Group D environments.

The standard CoolPoint flow rate transmitter is a 4-20 mA 3 wire transmitter, which has the following standard feature and user-configurable options: Brass body with viton seals and PVDF sensors; available in pipe sizes ranging from 1/4 inch to 4 inches; flow repeatability is +/- .25% (1/4 of 1%); +/- 2% full-scale accuracy at maximum flow; turndown ratio 10:1; and 3 wire 4-20 mA flow rate transmitter with visual 3-digit LED readout, selectable alarm state (N.O. or N.C.), set point or pulse output and engineering unit choice of GPM or LPM.

Universal Flow Monitors, www.flowmeters.com

Gas turbine meter

The Blancett Gas QuikSert Turbine Flow Meter provides Blancett's accuracy and ruggedness in a unique wafer-style design that allows for quick installation between two flanges. Gas QuikSert is constructed from stainless steel and tungsten carbide, ensuring an extended service life even in demanding oil and gas production environments. Explosion proof and intrinsically safe ratings allow QuikSert to be mounted in virtually all Class I Div 1 hazardous areas. With a lightweight rotor design that provides instantaneous response to changes in flow, and full compatibility with Blancett's B2800 Flow Monitors, K-Factor Scalers, Intelligent Converters, and the B3000 series of flow monitors, the Gas QuikSert provides an option for gas flow measurement up to 350 CFM [600 m3/H].

Racine Federated Inc., www.racinefed.com/flow

Handheld thermometer hygrometer data logger

The RHXL3SD Handheld Thermometer/Hygrometer Data Logger comes with a temperature-humidity probe, 2GB SD card, and features automatic temperature compensation for dew point, wet-bulb and thermocouple measurements, the ability to hold readings and display maximum and minimum readings, and an automatic power off function that extends battery life. The backlit dual LCD display makes readings easy to read. Ideal for clean rooms, greenhouses, and pharmaceutical industries to monitor the temperature and relative humidity.

Omega Engineering, www.omega.com

Test disconnect terminal blocks

The test disconnect terminal blocks save space in the control cabinet while increasing safety and reducing assembly time. The UTME 4 series features pluggable COMBI connectors in a compact 4 mm2 profile. The test disconnect blocks have short-circuit plugs for a "make first, break last" operation. This ensures the CT circuit is shorted out prior to test and measurement, preventing inadvertent actuation and improving safety for the system and its operators. The series' narrow profile saves cabinet space. A screw-free knife disconnect lever makes it easy to see the position of the switch, and there are bridging channels on either side of the disconnect lever.

Phoenix Contact, www.phoenixcontact.com

Windjammer brushless DC blowers

Windjammer brushless DC blowers offer compact, low-noise, and long-life solutions and includes low-voltage and high-voltage versions as well as bypass and thru-flow products. Sizes range from 3.0" to 5.7" to accommodate the most demanding design envelopes and application requirements. Low-voltage blowers (3.0", 3.3", 4.5", 5.0", 5.1", and 5.7") for vacuum or pressure applications can deliver variable output pressure up to 98 in. H2O and flows up to 190 CFM, depending on model, and high-voltage versions (5.7") can provide output pressure up to 169 in. H2O and flows up to 275 CFM. The blowers can accept voltage inputs of 12 VDC, 24 VDC, 48 VDC, 72 VDC, 120 VAC or 240 VAC.

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datafiles

Datafiles list useful literature on products and services that are available from manufacturers in the instrumentation and process-control industry. To receive free copies of this literature, please contact each manufacturer via their provided contact information.

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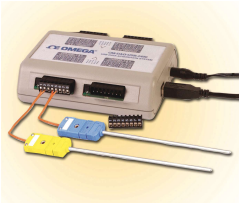

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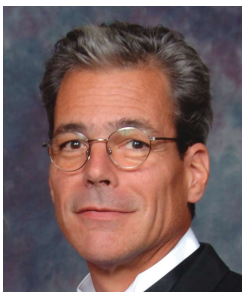
We're being acronymed and jargoned to death

By Paul Gruhn, P.E.



I realize the importance of acronyms and jargon. Both are shortcuts used to increase the efficiency of communications among experts. I am perfectly happy having the doctors and nurses in an emergency room using all the acronyms and jargon they want while they are patching me together. I really do not need to understand what they are saying, and it is in my best interest that they communicate as quickly and clearly as possible. Within the automation industry, we certainly have our own fair share of acronyms and jargon. However, business communications are meant to clarify, not obfuscate, so we should be careful about their use.

Most acronyms consist of three letters (e.g., PHA – Process Hazards Analysis). These are often jokingly referred to as TLAs – Three Letter Acronyms. Some acronyms are based on four letters (e.g., BPCS – Basic Process Control System). You might think these would be FLAs – Four Letter Acronyms – but you would be wrong. They're simply ETLAs – Extended Three Letter Acronyms. Some are based on five letters (e.g., HAZOP – HAZard and OPerability). Most five letter acronyms are based on a very selective choice of letters to make them roll off the tongue more easily. These are not FLAs either – Five Letter Acronyms. They are EETLAs – Enhanced Extended Three Letter Acronyms.

Within the automation industry, we certainly have our own fair share of acronyms and jargon. However, business communications are meant to clarify, not obfuscate, so we should be careful about their use.

The following are real acronyms taken from real documents. The additional comments on each, however, are my own.

- STEP – Sequential Timed Events Plotting. And I just thought steps were a part of stairs.
- CRIOP – Crisis Intervention and OPerability analysis. Another example of a selective choice of letters for an EETLA. After all, CIAOA would just sound like you are saying goodbye to someone in Italian.
- ICDE – International Common Cause Data Exchange. Isn't one of the Cs missing?

Can you believe some people actually use two letter acronyms? Where does one draw the line? Spelling out two words does not take up that much space! Perhaps we should call those TTLAs for Truncated Three Letter Acronyms.

- SO – Spurious Operation. So in case of a SO, we should do what?
- PC – Premature Closure. This sounds like a personal problem. Considering how the entire world uses that acronym for personal computer, I do not think this one will get much traction.
- UL – Utility Loss. I think Underwriter Laboratories might have a problem with this one.

Here is an actual sentence, although it is from a technical paper intended for a technical audience so I should not be too harsh. "If a SO does not lead to a spurious trip, for example, a single SIS element raising an alarm in a 2oo3 configuration, we may define the SO-failure as SD since the failure is detected before the SIF is executed." You got all that, right? Five acronyms in a single sentence, with one of them even repeated. SO is Spurious Operation, SIS is Safety Instrumented System, 2oo3 is Two out of Three (TooT wouldn't be very clear, would it?), SD is Safe Detected, and SIF is Safety Instrumented Function.

I have got a couple of new ones. How about SKIP for Safety Kernel Interrupt Procedure and LOU for Loss Of Utilities? SKIP to my LOU could now take on a whole new meaning!

For a hilarious treatment of buzzwords and jargon, check out some of the links and videos after Googling "retroencabulator and turboencabulator." Here is a partial excerpt. … "The original machine had a baseplate of prefabulated amulite, surmounted by a malleable logarithmic casing in such a way that the two spurving bearings were in a direct line with the pentametric fan. The main winding was of the normal lotus-o-delta type placed in panendermic semi-boloid slots in the stator, every seventh conductor being connected by a nonreversible trem'e pipe to the differential girdlespring on the 'up' end of the grammeters."

How many sales and marketing presentations have you been subjected to that were similar to that? Might you be guilty of creating any?

What the heck, if you can't dazzle 'em with brilliance …

ABOUT THE AUTHOR

Paul Gruhn, P.E., (paulg@icshou.com) works for ICS Triplex, a Rockwell Company. He is an ISA Fellow, standards committee member, book author, course developer/instructor, as well as a former Section, Division and Department leader for ISA. The views expressed here are his own and not those of his employer.

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